

Laboratory Report of Analysis

To: Kenai Watershed Forum
44129 Sterling Hwy
Soldotna, AK 99669

Report Number: **1243869**

Client Project: **Kenai Rive Baseline WQM**

Dear Benjamin Meyer,

Enclosed are the results of the analytical services performed under the referenced project for the received samples and associated QC as applicable. The samples are certified to meet the requirements of the National Environmental Laboratory Accreditation Conference Standards. Copies of this report and supporting data will be retained in our files for a period of ten years in the event they are required for future reference. All results are intended to be used in their entirety and SGS is not responsible for use of less than the complete report. Any samples submitted to our laboratory will be retained for a maximum of fourteen (14) days from the date of this report unless other archiving requirements were included in the quote.

If there are any questions about the report or services performed during this project, please call Curtis at (907) 562-2343. We will be happy to answer any questions or concerns which you may have.

Thank you for using SGS North America Inc. for your analytical services. We look forward to working with you again on any additional analytical needs.

Sincerely,
SGS North America Inc.

Curtis Whisman
Project Manager
curtis.whisman@sgs.com

Date

Case Narrative

SGS Client: **Kenai Watershed Forum**
 SGS Project: **1243869**
 Project Name/Site: **Kenai Rive Baseline WQM**
 Project Contact: **Benjamin Meyer**

Refer to sample receipt form for information on sample condition.

LCS for HBN 1895761 [MXX/36813 (1776584) LCS

200.8 - Metals LCS recovery for Thallium does not meet QC criteria (biased low).

MB for HBN 1895867 [MXX/36820] (1776876) MB

200.8 - Metals analyte Zinc is detected in the MB above the LOQ.

1243869029(1783042MS) (1776880) MS

200.8 - Metals MS recovery for Sodium does not meet QC criteria. Sample concentration is 4 times greater than the spike level.

1243862001MS (1777025) MS

4500NO3-F - Nitrate/Nitrite - MS recovery for Nitrite/Nitrate is outside of QC criteria. Refer to LCS for accuracy requirements.

1243862001MSD (1777026) MSD

4500NO3-F - Nitrate/Nitrite - MSD recovery for Nitrite/Nitrate is outside of QC criteria. Refer to LCS for accuracy requirements.

1243876001MS (1777027) MS

4500NO3-F - Nitrate/Nitrite - MS recovery for Nitrite/Nitrate is outside of QC criteria. Refer to LCS for accuracy requirements.

1243876001MSD (1777028) MSD

4500NO3-F - Nitrate/Nitrite - MSD recovery for Nitrite/Nitrate is outside of QC criteria. Refer to LCS for accuracy requirements.

QCS for HBN 1896238 (MMS/12372 (1777704) QCS

200.8 - Metals - QCS recovery for Thallium does not meet QC criteria. The associated sample concentrations are less than LOQ.

MB for HBN 1896339 [MXX/36834] (1778079) MB

200.8 - Metals analyte Zinc is detected in the MB above the LOQ. The associated sample concentrations are less than the LOQ.

QCS for HBN 1896574 (MMS/12377 (1778696) QCS

200.8 - Metals - QCS recovery for Aluminum does not meet QC criteria. The associated sample concentrations are less than LOQ.

*QC comments may be associated with the field samples found in this report. When applicable, comments will be applied to associated field samples.

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Laboratory Qualifiers

Enclosed are the analytical results associated with the above work order. The results apply to the samples as received. All results are intended to be used in their entirety and SGS is not responsible for use of less than the complete report. This document is issued by the Company under its General Conditions of Service accessible at <http://www.sgs.com/en/Terms-and-Conditions.aspx>. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein.

Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. Any unauthorized alteration, forgery or falsification of the context or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law.

SGS maintains a formal Quality Assurance/Quality Control (QA/QC) program. A copy of our Quality Assurance Plan (QAP), which outlines this program, is available at your request. The laboratory certification numbers are AK00971 (DW Chemistry & Microbiology) & 17-021 (CS) for ADEC and 2944.01 for DOD ELAP/ISO17025 (RCRA methods: 1020B, 1311, 3010A, 3050B, 3520C, 3550C, 5030B, 5035A, 6020B, 7470A, 7471B, 8015C, 8021B, 8082A, 8260D, 8270E, 8270E-SIM, 9040C, 9045D, 9056A, 9060A, AK101 and AK102/103). SGS is only certified for the analytes listed on our Drinking Water Certification (DW methods: 200.8, 2130B, 2320B, 2510B, 300.0, 4500-CN-C,E, 4500-H-B, 4500-NO3-F, 4500-P-E and 524.2) and only those analytes will be reported to the State of Alaska for compliance. Except as specifically noted, all statements and data in this report are in conformance to the provisions set forth by the SGS QAP and, when applicable, other regulatory authorities.

The following descriptors or qualifiers may be found in your report:

*	The analyte has exceeded allowable regulatory or control limits.
!	Surrogate out of control limits.
B	Indicates the analyte is found in a blank associated with the sample.
CCV/CVA/CVB	Continuing Calibration Verification
CCCV/CVC/CVCA/CVCB	Closing Continuing Calibration Verification
CL	Control Limit
DF	Analytical Dilution Factor
DL	Detection Limit (i.e., maximum method detection limit)
E	The analyte result is above the calibrated range.
GT	Greater Than
IB	Instrument Blank
ICV	Initial Calibration Verification
J	The quantitation is an estimation.
LCS(D)	Laboratory Control Spike (Duplicate)
LLQC/LLIQC	Low Level Quantitation Check
LOD	Limit of Detection (i.e., 3/4 of the LOQ)
LOQ	Limit of Quantitation (i.e., reporting or practical quantitation limit)
LT	Less Than
MB	Method Blank
MS(D)	Matrix Spike (Duplicate)
ND	Indicates the analyte is not detected.
RPD	Relative Percent Difference
TNTC	Too Numerous To Count
U	Indicates the analyte was analyzed for but not detected.

Note: Sample summaries which include a result for "Total Solids" have already been adjusted for moisture content. All DRO/RRO analyses are integrated per SOP.

Sample Summary

<u>Client Sample ID</u>	<u>Lab Sample ID</u>	<u>Collected</u>	<u>Received</u>	<u>Matrix</u>
RM 0 - No Name Creek	1243869001	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 1.5 - Kenai City Docks	1243869002	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 6.5 - Cunningham Park	1243869003	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 10 - Beaver Creek	1243869004	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 10.1 - Kenai River	1243869005	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 12.5 - Pillars	1243869006	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 18 - Poacher's Cove	1243869007	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 19 - Silkok Creek	1243869008	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 21 - Soldotna Bridge	1243869009	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 21 - Soldotna Bridge DUP	1243869010	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 22 - Soldotna Creek	1243869011	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 23 - Swiftwater Park	1243869012	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 30 - Funny River	1243869013	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 30 - Funny River DUP	1243869014	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 31 - Morgan's Landing	1243869015	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 36 - Moose River	1243869016	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 40 - Bing's Landing	1243869017	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 43 - Upstream of Dow Island	1243869018	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 44 - Mouth of Kiley River	1243869019	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 50 - Skilak Lake Outflow	1243869020	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 70 - Jim's Landing	1243869021	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 74 - Russian River	1243869022	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 82 - Kenai Lake Bridge	1243869023	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 79.5 - Juneau Creek	1243869024	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 1.5 - Trip Blank	1243869025	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 40, 43, 44, 50 - Trip Blank	1243869026	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 6.5 - Trip Blank	1243869027	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 0 - No Name Creek	1243869028	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 1.5 - Kenai City Docks	1243869029	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 6.5 - Cunningham Park	1243869030	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 10 - Beaver Creek	1243869031	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 10.1 - Kenai River	1243869032	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 12.5 - Pillars	1243869033	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 18 - Poacher's Cove	1243869034	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 19 - Silkok Creek	1243869035	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 21 - Soldotna Bridge	1243869036	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 21 - Soldotna Bridge DUP	1243869037	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 22 - Soldotna Creek	1243869038	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 23 - Swiftwater Park	1243869039	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)

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Sample Summary

<u>Client Sample ID</u>	<u>Lab Sample ID</u>	<u>Collected</u>	<u>Received</u>	<u>Matrix</u>
RM 30 - Funny River	1243869040	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 30 - Funny River DUP	1243869041	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 12.5 - Pillars	1243869042	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 22 - Soldotna Creek	1243869043	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 12.5 - Pillars	1243869044	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)
RM 22 - Soldotna Creek	1243869045	07/23/2024	07/23/2024	Water (Surface, Eff., Ground)

<u>Method</u>	<u>Method Description</u>
EP200.8	Metals in Drinking Water by ICP-MS DISSO
EP200.8	Metals in Water by 200.8 ICP-MS
SM21 4500NO3-F	Nitrate/Nitrite Flow injection Pres.
SM21 4500P-B,E	Total Phosphorus (W)
SW8260D	Volatile Organic Compounds (W)

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Detectable Results Summary

Client Sample ID: **RM 0 - No Name Creek**

Lab Sample ID: 1243869001

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	13300	ug/L
Iron	3240	ug/L
Magnesium	5800	ug/L
Zinc	5.24J	ug/L
Total Nitrate/Nitrite-N	0.0842J	mg/L

Waters Department

Client Sample ID: **RM 1.5 - Kenai City Docks**

Lab Sample ID: 1243869002

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	35900	ug/L
Copper	17.1	ug/L
Iron	8690	ug/L
Magnesium	67900	ug/L
Zinc	30.9	ug/L
Total Nitrate/Nitrite-N	0.234	mg/L
Total Phosphorus	0.210	mg/L

Waters Department

Client Sample ID: **RM 6.5 - Cunningham Park**

Lab Sample ID: 1243869003

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	12600	ug/L
Copper	1.64J	ug/L
Iron	810	ug/L
Magnesium	1460	ug/L
Zinc	4.79J	ug/L
Total Nitrate/Nitrite-N	0.212	mg/L
Total Phosphorus	0.0122J	mg/L

Waters Department

Client Sample ID: **RM 10 - Beaver Creek**

Lab Sample ID: 1243869004

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	17200	ug/L
Iron	2480	ug/L
Magnesium	4250	ug/L
Total Nitrate/Nitrite-N	0.0724J	mg/L
Total Phosphorus	0.0589	mg/L

Waters Department

Client Sample ID: **RM 10.1 - Kenai River**

Lab Sample ID: 1243869005

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	13600	ug/L
Copper	1.12J	ug/L
Iron	516	ug/L
Magnesium	1430	ug/L
Zinc	3.22J	ug/L
Total Nitrate/Nitrite-N	0.222	mg/L

Waters Department

Detectable Results Summary

Client Sample ID: **RM 12.5 - Pillars**

Lab Sample ID: 1243869006

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	13000	ug/L
Copper	1.11J	ug/L
Iron	300	ug/L
Magnesium	1250	ug/L
Total Nitrate/Nitrite-N	0.210	mg/L

Waters Department

Client Sample ID: **RM 18 - Poacher's Cove**

Lab Sample ID: 1243869007

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	13400	ug/L
Iron	249J	ug/L
Magnesium	1260	ug/L
Zinc	3.11J	ug/L
Total Nitrate/Nitrite-N	0.229	mg/L

Waters Department

Client Sample ID: **RM 19 - Silkok Creek**

Lab Sample ID: 1243869008

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	18000	ug/L
Iron	1780	ug/L
Magnesium	5270	ug/L
Zinc	23.0	ug/L
Total Nitrate/Nitrite-N	0.189J	mg/L

Waters Department

Client Sample ID: **RM 21 - Soldotna Bridge**

Lab Sample ID: 1243869009

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	12700	ug/L
Iron	257	ug/L
Magnesium	1180	ug/L
Zinc	10.9	ug/L
Total Nitrate/Nitrite-N	0.201	mg/L

Waters Department

Client Sample ID: **RM 21 - Soldotna Bridge DUP**

Lab Sample ID: 1243869010

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	12800	ug/L
Iron	279	ug/L
Magnesium	1160	ug/L
Zinc	12.4	ug/L
Total Nitrate/Nitrite-N	0.195J	mg/L

Waters Department

Client Sample ID: **RM 22 - Soldotna Creek**

Lab Sample ID: 1243869011

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	21300	ug/L
Iron	895	ug/L
Magnesium	6020	ug/L
Zinc	16.9	ug/L
Total Nitrate/Nitrite-N	0.0660J	mg/L
Total Phosphorus	0.0749	mg/L

Waters Department

Detectable Results Summary

Client Sample ID: **RM 23 - Swiftwater Park**

Lab Sample ID: 1243869012

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	13200	ug/L
Iron	286	ug/L
Magnesium	1220	ug/L
Zinc	13.4	ug/L
Total Nitrate/Nitrite-N	0.246	mg/L

Waters Department

Client Sample ID: **RM 30 - Funny River**

Lab Sample ID: 1243869013

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	11300	ug/L
Copper	1.17J	ug/L
Iron	1410	ug/L
Magnesium	3810	ug/L
Zinc	12.2	ug/L
Total Phosphorus	0.0610	mg/L

Waters Department

Client Sample ID: **RM 30 - Funny River DUP**

Lab Sample ID: 1243869014

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	10500	ug/L
Iron	1030	ug/L
Magnesium	3460	ug/L
Zinc	13.6	ug/L
Total Phosphorus	0.0286J	mg/L

Waters Department

Client Sample ID: **RM 31 - Morgan's Landing**

Lab Sample ID: 1243869015

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	12900	ug/L
Iron	351	ug/L
Magnesium	1170	ug/L
Zinc	12.6	ug/L
Total Nitrate/Nitrite-N	0.211	mg/L

Waters Department

Client Sample ID: **RM 40 - Bing's Landing**

Lab Sample ID: 1243869017

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	12000	ug/L
Iron	282	ug/L
Magnesium	1060	ug/L
Total Nitrate/Nitrite-N	0.295	mg/L

Waters Department

Client Sample ID: **RM 43 - Upstream of Dow Island**

Lab Sample ID: 1243869018

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	12600	ug/L
Iron	342	ug/L
Magnesium	1110	ug/L
Total Nitrate/Nitrite-N	0.230	mg/L

Waters Department

Detectable Results Summary

Client Sample ID: **RM 44 - Mouth of Kiley River**

Lab Sample ID: 1243869019

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	8310	ug/L
Iron	1300	ug/L
Magnesium	1450	ug/L
Total Nitrate/Nitrite-N	0.110J	mg/L

Waters Department

Client Sample ID: **RM 50 - Skilak Lake Outflow**

Lab Sample ID: 1243869020

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	13100	ug/L
Magnesium	1030	ug/L
Total Nitrate/Nitrite-N	0.254	mg/L

Waters Department

Client Sample ID: **RM 70 - Jim's Landing**

Lab Sample ID: 1243869021

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	16700	ug/L
Magnesium	1370	ug/L
Total Nitrate/Nitrite-N	0.333	mg/L

Waters Department

Client Sample ID: **RM 74 - Russian River**

Lab Sample ID: 1243869022

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	18600	ug/L
Magnesium	1230	ug/L
Total Nitrate/Nitrite-N	0.520	mg/L

Waters Department

Client Sample ID: **RM 82 - Kenai Lake Bridge**

Lab Sample ID: 1243869023

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	16700	ug/L
Magnesium	1350	ug/L
Total Nitrate/Nitrite-N	0.302	mg/L

Waters Department

Client Sample ID: **RM 79.5 - Juneau Creek**

Lab Sample ID: 1243869024

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Calcium	16300	ug/L
Magnesium	1240	ug/L
Total Nitrate/Nitrite-N	0.156J	mg/L

Waters Department

Client Sample ID: **RM 0 - No Name Creek**

Lab Sample ID: 1243869028

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	16.5	ug/L

Client Sample ID: **RM 1.5 - Kenai City Docks**

Lab Sample ID: 1243869029

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Arsenic	1.84J	ug/L
Copper	5.57	ug/L
Zinc	16.8	ug/L

Detectable Results Summary

Client Sample ID: **RM 6.5 - Cunningham Park**

Lab Sample ID: 1243869030

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Arsenic	1.56J	ug/L
Chromium	3.27J	ug/L
Copper	1.28J	ug/L
Zinc	19.2	ug/L

Client Sample ID: **RM 10 - Beaver Creek**

Lab Sample ID: 1243869031

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	8.48J	ug/L

Client Sample ID: **RM 10.1 - Kenai River**

Lab Sample ID: 1243869032

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Copper	1.09J	ug/L
Zinc	8.76J	ug/L

Client Sample ID: **RM 12.5 - Pillars**

Lab Sample ID: 1243869033

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Copper	1.47J	ug/L
Zinc	10.5	ug/L

Client Sample ID: **RM 18 - Poacher's Cove**

Lab Sample ID: 1243869034

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Lead	0.589J	ug/L
Zinc	9.41J	ug/L

Client Sample ID: **RM 19 - Silkok Creek**

Lab Sample ID: 1243869035

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Chromium	6.74	ug/L
Zinc	6.96J	ug/L

Client Sample ID: **RM 21 - Soldotna Bridge**

Lab Sample ID: 1243869036

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Chromium	2.72J	ug/L
Zinc	8.21J	ug/L

Client Sample ID: **RM 21 - Soldotna Bridge DUP**

Lab Sample ID: 1243869037

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	8.23J	ug/L

Client Sample ID: **RM 22 - Soldotna Creek**

Lab Sample ID: 1243869038

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Chromium	3.62J	ug/L
Zinc	9.52J	ug/L

Client Sample ID: **RM 23 - Swiftwater Park**

Lab Sample ID: 1243869039

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	8.54J	ug/L

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Member of SGS Group

Detectable Results Summary

Client Sample ID: **RM 30 - Funny River**

Lab Sample ID: 1243869040

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	10.0	ug/L

Client Sample ID: **RM 30 - Funny River DUP**

Lab Sample ID: 1243869041

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	7.71J	ug/L

Client Sample ID: **RM 12.5 - Pillars**

Lab Sample ID: 1243869042

Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Chromium	8.33	ug/L

Client Sample ID: **RM 12.5 - Pillars**

Lab Sample ID: 1243869044

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Zinc	8.74J	ug/L

Client Sample ID: **RM 22 - Soldotna Creek**

Lab Sample ID: 1243869045

Dissolved Metals by ICP/MS

<u>Parameter</u>	<u>Result</u>	<u>Units</u>
Arsenic	4.20J	ug/L
Chromium	4.48J	ug/L
Zinc	7.99J	ug/L

Results of RM 0 - No Name Creek

Client Sample ID: **RM 0 - No Name Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869001
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:59
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	13300		500	150	375	ug/L	1		07/29/24 22:39
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 22:39
Iron	3240		250	78.0	188	ug/L	1		07/29/24 22:39
Magnesium	5800		50.0	15.0	37.5	ug/L	1		07/29/24 22:39
Zinc	5.24	J	10.0	3.10	7.50	ug/L	1		07/29/24 22:39

Batch Information

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 22:39
 Container ID: 1243869001-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 0 - No Name Creek

Client Sample ID: **RM 0 - No Name Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869001
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:59
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable</u> <u>Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.0842	J	0.200	0.0500	0.150	mg/L	2		07/26/24 13:53

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 13:53
 Container ID: 1243869001-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable</u> <u>Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:12

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:12
 Container ID: 1243869001-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 1.5 - Kenai City Docks

Client Sample ID: **RM 1.5 - Kenai City Docks**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869002
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	35900		2500	750	1875	ug/L	5		07/30/24 13:04
Copper	17.1		3.00	1.00	2.25	ug/L	1		07/29/24 22:42
Iron	8690		250	78.0	188	ug/L	1		07/29/24 22:42
Magnesium	67900		250	75.0	188	ug/L	5		07/30/24 13:04
Zinc	30.9		10.0	3.10	7.50	ug/L	1		07/29/24 22:42

Batch Information

Analytical Batch: MMS12372
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/30/24 13:04
 Container ID: 1243869002-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 22:42
 Container ID: 1243869002-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 1.5 - Kenai City Docks

Client Sample ID: **RM 1.5 - Kenai City Docks**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869002
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 08:01
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:01
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:01
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 08:01
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:01
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 08:01

Surrogates

1,2-Dichloroethane-D4 (surr)	101		81-118			%	1		08/03/24 08:01
4-Bromofluorobenzene (surr)	101		85-114			%	1		08/03/24 08:01
Toluene-d8 (surr)	102		89-112			%	1		08/03/24 08:01

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 08:01
 Container ID: 1243869002-C

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 1.5 - Kenai City Docks

Client Sample ID: **RM 1.5 - Kenai City Docks**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869002
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.234		0.200	0.0500	0.150	mg/L	2		07/26/24 13:54

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 13:54
 Container ID: 1243869002-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.210		0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:15

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:15
 Container ID: 1243869002-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 6.5 - Cunningham Park

Client Sample ID: **RM 6.5 - Cunningham Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869003
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:20
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	12600		500	150	375	ug/L	1		07/29/24 22:44
Copper	1.64	J	3.00	1.00	2.25	ug/L	1		07/29/24 22:44
Iron	810		250	78.0	188	ug/L	1		07/29/24 22:44
Magnesium	1460		50.0	15.0	37.5	ug/L	1		07/29/24 22:44
Zinc	4.79	J	10.0	3.10	7.50	ug/L	1		07/29/24 22:44

Batch Information

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 22:44
 Container ID: 1243869003-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 6.5 - Cunningham Park

Client Sample ID: **RM 6.5 - Cunningham Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869003
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:20
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 08:31
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:31
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:31
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 08:31
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:31
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 08:31

Surrogates

1,2-Dichloroethane-D4 (surr)	101		81-118			%	1		08/03/24 08:31
4-Bromofluorobenzene (surr)	100		85-114			%	1		08/03/24 08:31
Toluene-d8 (surr)	102		89-112			%	1		08/03/24 08:31

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 08:31
 Container ID: 1243869003-C

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 6.5 - Cunningham Park

Client Sample ID: **RM 6.5 - Cunningham Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869003
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:20
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.212		0.200	0.0500	0.150	mg/L	2		07/26/24 13:56

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 13:56
 Container ID: 1243869003-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0122	J	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:18

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:18
 Container ID: 1243869003-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 10 - Beaver Creek

Client Sample ID: **RM 10 - Beaver Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869004
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:05
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	17200		500	150	375	ug/L	1		07/29/24 22:47
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 22:47
Iron	2480		250	78.0	188	ug/L	1		07/29/24 22:47
Magnesium	4250		50.0	15.0	37.5	ug/L	1		07/29/24 22:47
Zinc	7.50	U	10.0	3.10	7.50	ug/L	1		07/29/24 22:47

Batch Information

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 22:47
 Container ID: 1243869004-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 10 - Beaver Creek

Client Sample ID: **RM 10 - Beaver Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869004
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:05
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.0724	J	0.200	0.0500	0.150	mg/L	2		07/26/24 13:58

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 13:58
 Container ID: 1243869004-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0589		0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:19

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:19
 Container ID: 1243869004-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 10.1 - Kenai River

Client Sample ID: **RM 10.1 - Kenai River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869005
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:32
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	13600		500	150	375	ug/L	1		07/29/24 22:55
Copper	1.12	J	3.00	1.00	2.25	ug/L	1		07/29/24 22:55
Iron	516		250	78.0	188	ug/L	1		07/29/24 22:55
Magnesium	1430		50.0	15.0	37.5	ug/L	1		07/29/24 22:55
Zinc	3.22	J	10.0	3.10	7.50	ug/L	1		07/29/24 22:55

Batch Information

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 22:55
 Container ID: 1243869005-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 10.1 - Kenai River

Client Sample ID: **RM 10.1 - Kenai River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869005
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:32
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.222		0.200	0.0500	0.150	mg/L	2		07/26/24 14:00

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:00
 Container ID: 1243869005-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:20

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:20
 Container ID: 1243869005-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 12.5 - Pillars

Client Sample ID: **RM 12.5 - Pillars**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869006
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	13000		500	150	375	ug/L	1		07/29/24 22:58
Copper	1.11	J	3.00	1.00	2.25	ug/L	1		07/29/24 22:58
Iron	300		250	78.0	188	ug/L	1		07/29/24 22:58
Magnesium	1250		50.0	15.0	37.5	ug/L	1		07/29/24 22:58
Zinc	7.50	U	10.0	3.10	7.50	ug/L	1		07/29/24 22:58

Batch Information

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 22:58
 Container ID: 1243869006-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 12.5 - Pillars

Client Sample ID: **RM 12.5 - Pillars**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869006
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.210		0.200	0.0500	0.150	mg/L	2		07/26/24 14:01

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:01
 Container ID: 1243869006-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:21

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:21
 Container ID: 1243869006-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 18 - Poacher's Cove

Client Sample ID: **RM 18 - Poacher's Cove**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869007
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:37
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	13400		500	150	375	ug/L	1		07/29/24 23:01
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 23:01
Iron	249	J	250	78.0	188	ug/L	1		07/29/24 23:01
Magnesium	1260		50.0	15.0	37.5	ug/L	1		07/29/24 23:01
Zinc	3.11	J	10.0	3.10	7.50	ug/L	1		07/29/24 23:01

Batch Information

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 23:01
 Container ID: 1243869007-B

Prep Batch: MXX36813
 Prep Method: E200.2
 Prep Date/Time: 07/25/24 10:45
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 18 - Poacher's Cove

Client Sample ID: **RM 18 - Poacher's Cove**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869007
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:37
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.229		0.200	0.0500	0.150	mg/L	2		07/26/24 14:03

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:03
 Container ID: 1243869007-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:22

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:22
 Container ID: 1243869007-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 19 - Silkok Creek

Client Sample ID: **RM 19 - Silkok Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869008
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:32
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	18000		500	150	375	ug/L	1		08/02/24 16:18
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:18
Iron	1780		250	78.0	188	ug/L	1		08/02/24 16:18
Magnesium	5270		50.0	15.0	37.5	ug/L	1		08/02/24 16:18
Zinc	23.0	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:18

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:18
 Container ID: 1243869008-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 19 - Silkok Creek

Client Sample ID: **RM 19 - Silkok Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869008
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:32
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.189	J	0.200	0.0500	0.150	mg/L	2		07/26/24 14:15

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:15
 Container ID: 1243869008-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:23

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:23
 Container ID: 1243869008-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 21 - Soldotna Bridge

Client Sample ID: **RM 21 - Soldotna Bridge**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869009
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:03
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	12700		500	150	375	ug/L	1		08/02/24 16:20
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:20
Iron	257		250	78.0	188	ug/L	1		08/02/24 16:20
Magnesium	1180		50.0	15.0	37.5	ug/L	1		08/02/24 16:20
Zinc	10.9	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:20

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:20
 Container ID: 1243869009-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 21 - Soldotna Bridge

Client Sample ID: **RM 21 - Soldotna Bridge**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869009
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:03
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.201		0.200	0.0500	0.150	mg/L	2		07/26/24 14:17

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:17
 Container ID: 1243869009-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:24

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:24
 Container ID: 1243869009-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 21 - Soldotna Bridge DUP

Client Sample ID: **RM 21 - Soldotna Bridge DUP**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869010
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:09
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	12800		500	150	375	ug/L	1		08/02/24 16:23
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:23
Iron	279		250	78.0	188	ug/L	1		08/02/24 16:23
Magnesium	1160		50.0	15.0	37.5	ug/L	1		08/02/24 16:23
Zinc	12.4	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:23

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:23
 Container ID: 1243869010-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 21 - Soldotna Bridge DUP

Client Sample ID: **RM 21 - Soldotna Bridge DUP**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869010
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:09
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.195	J	0.200	0.0500	0.150	mg/L	2		07/26/24 14:19

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:19
 Container ID: 1243869010-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:25

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:25
 Container ID: 1243869010-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 22 - Soldotna Creek

Client Sample ID: **RM 22 - Soldotna Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869011
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:45
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	21300		500	150	375	ug/L	1		08/02/24 16:25
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:25
Iron	895		250	78.0	188	ug/L	1		08/02/24 16:25
Magnesium	6020		50.0	15.0	37.5	ug/L	1		08/02/24 16:25
Zinc	16.9	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:25

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:25
 Container ID: 1243869011-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 22 - Soldotna Creek

Client Sample ID: **RM 22 - Soldotna Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869011
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:45
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.0660	J	0.200	0.0500	0.150	mg/L	2		07/26/24 14:21

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:21
 Container ID: 1243869011-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0749		0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:26

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:26
 Container ID: 1243869011-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 23 - Swiftwater Park

Client Sample ID: **RM 23 - Swiftwater Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869012
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:30
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	13200		500	150	375	ug/L	1		08/02/24 16:28
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:28
Iron	286		250	78.0	188	ug/L	1		08/02/24 16:28
Magnesium	1220		50.0	15.0	37.5	ug/L	1		08/02/24 16:28
Zinc	13.4	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:28

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:28
 Container ID: 1243869012-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 23 - Swiftwater Park

Client Sample ID: **RM 23 - Swiftwater Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869012
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:30
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable</u> <u>Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.246		0.200	0.0500	0.150	mg/L	2		07/26/24 14:22

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:22
 Container ID: 1243869012-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable</u> <u>Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:26

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:26
 Container ID: 1243869012-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 30 - Funny River

Client Sample ID: **RM 30 - Funny River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869013
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:52
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	11300		500	150	375	ug/L	1		08/02/24 16:30
Copper	1.17	J	3.00	1.00	2.25	ug/L	1		08/02/24 16:30
Iron	1410		250	78.0	188	ug/L	1		08/02/24 16:30
Magnesium	3810		50.0	15.0	37.5	ug/L	1		08/02/24 16:30
Zinc	12.2	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:30

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:30
 Container ID: 1243869013-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 30 - Funny River

Client Sample ID: **RM 30 - Funny River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869013
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:52
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.150	U	0.200	0.0500	0.150	mg/L	2		07/26/24 14:24

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:24
 Container ID: 1243869013-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0610		0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:29

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:29
 Container ID: 1243869013-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 30 - Funny River DUP

Client Sample ID: **RM 30 - Funny River DUP**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869014
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:10
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	10500		500	150	375	ug/L	1		08/02/24 16:33
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:33
Iron	1030		250	78.0	188	ug/L	1		08/02/24 16:33
Magnesium	3460		50.0	15.0	37.5	ug/L	1		08/02/24 16:33
Zinc	13.6	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:33

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:33
 Container ID: 1243869014-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 30 - Funny River DUP

Client Sample ID: **RM 30 - Funny River DUP**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869014
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:10
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.150	U	0.200	0.0500	0.150	mg/L	2		07/26/24 14:26

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:26
 Container ID: 1243869014-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0286	J	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:30

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:30
 Container ID: 1243869014-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 31 - Morgan's Landing

Client Sample ID: **RM 31 - Morgan's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869015
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:23
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	12900		500	150	375	ug/L	1		08/02/24 16:41
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 16:41
Iron	351		250	78.0	188	ug/L	1		08/02/24 16:41
Magnesium	1170		50.0	15.0	37.5	ug/L	1		08/02/24 16:41
Zinc	12.6	B	10.0	3.10	7.50	ug/L	1		08/02/24 16:41

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:41
 Container ID: 1243869015-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 31 - Morgan's Landing

Client Sample ID: **RM 31 - Morgan's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869015
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:23
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.211		0.200	0.0500	0.150	mg/L	2		07/26/24 14:33

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:33
 Container ID: 1243869015-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:31

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:31
 Container ID: 1243869015-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 36 - Moose River

Client Sample ID: **RM 36 - Moose River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869016
 Lab Project ID: 1243869

Collection Date: 07/23/24 12:40
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.150	U	0.200	0.0500	0.150	mg/L	2		07/26/24 14:35

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:35
 Container ID: 1243869016-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:32

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:32
 Container ID: 1243869016-A

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 40 - Bing's Landing

Client Sample ID: **RM 40 - Bing's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869017
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	12000		500	150	375	ug/L	1		08/02/24 16:43
Iron	282		250	78.0	188	ug/L	1		08/02/24 16:43
Magnesium	1060		50.0	15.0	37.5	ug/L	1		08/02/24 16:43

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:43
 Container ID: 1243869017-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 40 - Bing's Landing

Client Sample ID: **RM 40 - Bing's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869017
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 07:46
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 07:46
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 07:46
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 07:46
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 07:46
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 07:46

Surrogates

1,2-Dichloroethane-D4 (surr)	102		81-118			%	1		08/03/24 07:46
4-Bromofluorobenzene (surr)	99.9		85-114			%	1		08/03/24 07:46
Toluene-d8 (surr)	103		89-112			%	1		08/03/24 07:46

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 07:46
 Container ID: 1243869017-C

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 40 - Bing's Landing

Client Sample ID: **RM 40 - Bing's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869017
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.295		0.200	0.0500	0.150	mg/L	2		07/26/24 14:36

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:36
 Container ID: 1243869017-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:36

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:36
 Container ID: 1243869017-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 43 - Upstream of Dow Island

Client Sample ID: **RM 43 - Upstream of Dow Island**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869018
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:25
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	12600		500	150	375	ug/L	1		08/02/24 16:46
Iron	342		250	78.0	188	ug/L	1		08/02/24 16:46
Magnesium	1110		50.0	15.0	37.5	ug/L	1		08/02/24 16:46

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:46
 Container ID: 1243869018-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 43 - Upstream of Dow Island

Client Sample ID: **RM 43 - Upstream of Dow Island**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869018
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:25
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 08:16
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:16
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:16
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 08:16
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 08:16
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 08:16

Surrogates

1,2-Dichloroethane-D4 (surr)	99.9		81-118			%	1		08/03/24 08:16
4-Bromofluorobenzene (surr)	101		85-114			%	1		08/03/24 08:16
Toluene-d8 (surr)	102		89-112			%	1		08/03/24 08:16

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 08:16
 Container ID: 1243869018-C

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 43 - Upstream of Dow Island

Client Sample ID: **RM 43 - Upstream of Dow Island**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869018
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:25
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.230		0.200	0.0500	0.150	mg/L	2		07/26/24 14:38

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:38
 Container ID: 1243869018-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:41

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:41
 Container ID: 1243869018-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 44 - Mouth of Kiley River

Client Sample ID: **RM 44 - Mouth of Kiley River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869019
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	8310		500	150	375	ug/L	1		08/02/24 16:48
Iron	1300		250	78.0	188	ug/L	1		08/02/24 16:48
Magnesium	1450		50.0	15.0	37.5	ug/L	1		08/02/24 16:48

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:48
 Container ID: 1243869019-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 44 - Mouth of Kiley River

Client Sample ID: **RM 44 - Mouth of Kiley River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869019
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.110	J	0.200	0.0500	0.150	mg/L	2		07/26/24 14:40

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:40
 Container ID: 1243869019-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:41

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:41
 Container ID: 1243869019-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 50 - Skilak Lake Outflow

Client Sample ID: **RM 50 - Skilak Lake Outflow**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869020
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:05
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	13100		500	150	375	ug/L	1		08/02/24 16:51
Iron	188	U	250	78.0	188	ug/L	1		08/02/24 16:51
Magnesium	1030		50.0	15.0	37.5	ug/L	1		08/02/24 16:51

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:51
 Container ID: 1243869020-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 50 - Skilak Lake Outflow

Client Sample ID: **RM 50 - Skilak Lake Outflow**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869020
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:05
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.254		0.200	0.0500	0.150	mg/L	2		07/26/24 14:42

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:42
 Container ID: 1243869020-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:42

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:42
 Container ID: 1243869020-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 70 - Jim's Landing

Client Sample ID: **RM 70 - Jim's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869021
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:50
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	16700		500	150	375	ug/L	1		08/02/24 16:54
Iron	188	U	250	78.0	188	ug/L	1		08/02/24 16:54
Magnesium	1370		50.0	15.0	37.5	ug/L	1		08/02/24 16:54

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:54
 Container ID: 1243869021-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 70 - Jim's Landing

Client Sample ID: **RM 70 - Jim's Landing**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869021
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:50
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.333		0.200	0.0500	0.150	mg/L	2		07/26/24 14:43

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:43
 Container ID: 1243869021-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:43

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:43
 Container ID: 1243869021-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 74 - Russian River

Client Sample ID: **RM 74 - Russian River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869022
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	18600		500	150	375	ug/L	1		08/02/24 16:56
Iron	188	U	250	78.0	188	ug/L	1		08/02/24 16:56
Magnesium	1230		50.0	15.0	37.5	ug/L	1		08/02/24 16:56

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:56
 Container ID: 1243869022-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 74 - Russian River

Client Sample ID: **RM 74 - Russian River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869022
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.520		0.200	0.0500	0.150	mg/L	2		07/26/24 14:45

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:45
 Container ID: 1243869022-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:44

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:44
 Container ID: 1243869022-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 82 - Kenai Lake Bridge

Client Sample ID: **RM 82 - Kenai Lake Bridge**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869023
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:40
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	16700		500	150	375	ug/L	1		08/02/24 16:59
Iron	188	U	250	78.0	188	ug/L	1		08/02/24 16:59
Magnesium	1350		50.0	15.0	37.5	ug/L	1		08/02/24 16:59

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 16:59
 Container ID: 1243869023-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 82 - Kenai Lake Bridge

Client Sample ID: **RM 82 - Kenai Lake Bridge**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869023
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:40
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Nitrate/Nitrite-N	0.302		0.200	0.0500	0.150	mg/L	2		07/26/24 14:47

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:47
 Container ID: 1243869023-A

<u>Parameter</u>	<u>Result</u>	<u>Qual</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>	<u>DF</u>	<u>Allowable Limits</u>	<u>Date Analyzed</u>
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:45

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:45
 Container ID: 1243869023-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 79.5 - Juneau Creek

Client Sample ID: **RM 79.5 - Juneau Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869024
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:40
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Calcium	16300		500	150	375	ug/L	1		08/02/24 17:02
Iron	188	U	250	78.0	188	ug/L	1		08/02/24 17:02
Magnesium	1240		50.0	15.0	37.5	ug/L	1		08/02/24 17:02

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:02
 Container ID: 1243869024-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 79.5 - Juneau Creek

Client Sample ID: **RM 79.5 - Juneau Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869024
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:40
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Waters Department

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Nitrate/Nitrite-N	0.156	J	0.200	0.0500	0.150	mg/L	2		07/26/24 14:49

Batch Information

Analytical Batch: WFI3132
 Analytical Method: SM21 4500NO3-F
 Analyst: AJP
 Analytical Date/Time: 07/26/24 14:49
 Container ID: 1243869024-A

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Total Phosphorus	0.0300	U	0.0400	0.0120	0.0300	mg/L	1		08/01/24 13:46

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Analyst: AJP
 Analytical Date/Time: 08/01/24 13:46
 Container ID: 1243869024-A

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/24 12:14
 Prep Initial Wt./Vol.: 25 mL
 Prep Extract Vol: 25 mL

Results of RM 1.5 - Trip Blank

Client Sample ID: **RM 1.5 - Trip Blank**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869025
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 05:58
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 05:58
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 05:58
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 05:58
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 05:58
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 05:58

Surrogates

1,2-Dichloroethane-D4 (surr)	100		81-118			%	1		08/03/24 05:58
4-Bromofluorobenzene (surr)	99.2		85-114			%	1		08/03/24 05:58
Toluene-d8 (surr)	102		89-112			%	1		08/03/24 05:58

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 05:58
 Container ID: 1243869025-A

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 40, 43, 44, 50 - Trip Blank

Client Sample ID: **RM 40, 43, 44, 50 - Trip Blank**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869026
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:25
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 06:13
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 06:13
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 06:13
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 06:13
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 06:13
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 06:13

Surrogates

1,2-Dichloroethane-D4 (surr)	102		81-118			%	1		08/03/24 06:13
4-Bromofluorobenzene (surr)	99.2		85-114			%	1		08/03/24 06:13
Toluene-d8 (surr)	101		89-112			%	1		08/03/24 06:13

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 06:13
 Container ID: 1243869026-A

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 6.5 - Trip Blank

Client Sample ID: **RM 6.5 - Trip Blank**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869027
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:20
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Volatile GC/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Benzene	0.300	U	0.400	0.120	0.300	ug/L	1		08/03/24 06:28
Ethylbenzene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 06:28
o-Xylene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 06:28
P & M -Xylene	1.50	U	2.00	0.620	1.50	ug/L	1		08/03/24 06:28
Toluene	0.750	U	1.00	0.310	0.750	ug/L	1		08/03/24 06:28
Xylenes (total)	2.25	U	3.00	1.00	2.25	ug/L	1		08/03/24 06:28

Surrogates

1,2-Dichloroethane-D4 (surr)	102		81-118			%	1		08/03/24 06:28
4-Bromofluorobenzene (surr)	102		85-114			%	1		08/03/24 06:28
Toluene-d8 (surr)	102		89-112			%	1		08/03/24 06:28

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Analyst: MZ
 Analytical Date/Time: 08/03/24 06:28
 Container ID: 1243869027-A

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/24 06:00
 Prep Initial Wt./Vol.: 5 mL
 Prep Extract Vol: 5 mL

Results of RM 0 - No Name Creek

Client Sample ID: **RM 0 - No Name Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869028
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:59
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		08/02/24 17:04
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		08/02/24 17:04
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		08/02/24 17:04
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		08/02/24 17:04
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		08/02/24 17:04
Zinc	16.5	B	10.0	3.10	7.50	ug/L	1		08/02/24 17:04

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:04
 Container ID: 1243869028-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 1.5 - Kenai City Docks

Client Sample ID: **RM 1.5 - Kenai City Docks**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869029
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:00
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	1.84	J	5.00	1.50	3.75	ug/L	1		08/02/24 17:14
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		08/02/24 17:14
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		08/02/24 17:14
Copper	5.57		3.00	1.00	2.25	ug/L	1		08/02/24 17:14
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		08/02/24 17:14
Zinc	16.8	B	10.0	3.10	7.50	ug/L	1		08/02/24 17:14

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:14
 Container ID: 1243869029-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 6.5 - Cunningham Park

Client Sample ID: **RM 6.5 - Cunningham Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869030
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:20
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	1.56	J	5.00	1.50	3.75	ug/L	1		08/02/24 17:18
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		08/02/24 17:18
Chromium	3.27	J	5.00	2.50	3.75	ug/L	1		08/02/24 17:18
Copper	1.28	J	3.00	1.00	2.25	ug/L	1		08/02/24 17:18
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		08/02/24 17:18
Zinc	19.2	B	10.0	3.10	7.50	ug/L	1		08/02/24 17:18

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:18
 Container ID: 1243869030-B

Prep Batch: MXX36820
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 10 - Beaver Creek

Client Sample ID: **RM 10 - Beaver Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869031
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:05
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 19:44
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 19:44
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 19:44
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 19:44
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 19:44
Zinc	8.48	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:28

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:28
 Container ID: 1243869031-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 19:44
 Container ID: 1243869031-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 10.1 - Kenai River

Client Sample ID: **RM 10.1 - Kenai River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869032
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:32
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 19:53
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 19:53
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 19:53
Copper	1.09	J	3.00	1.00	2.25	ug/L	1		07/29/24 19:53
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 19:53
Zinc	8.76	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:31

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:31
 Container ID: 1243869032-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 19:53
 Container ID: 1243869032-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 12.5 - Pillars

Client Sample ID: **RM 12.5 - Pillars**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869033
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 19:58
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 19:58
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 19:58
Copper	1.47	J	3.00	1.00	2.25	ug/L	1		07/29/24 19:58
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 19:58
Zinc	10.5	B	10.0	3.10	7.50	ug/L	1		08/02/24 17:33

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:33
 Container ID: 1243869033-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 19:58
 Container ID: 1243869033-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 18 - Poacher's Cove

Client Sample ID: **RM 18 - Poacher's Cove**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869034
 Lab Project ID: 1243869

Collection Date: 07/23/24 11:37
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:00
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:00
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:00
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:00
Lead	0.589	J	2.00	0.500	1.50	ug/L	1		07/29/24 20:00
Zinc	9.41	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:41

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:41
 Container ID: 1243869034-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:00
 Container ID: 1243869034-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 19 - Silkok Creek

Client Sample ID: **RM 19 - Silkok Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869035
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:32
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:03
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:03
Chromium	6.74		5.00	2.50	3.75	ug/L	1		07/29/24 20:03
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:03
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:03
Zinc	6.96	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:46

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:46
 Container ID: 1243869035-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:03
 Container ID: 1243869035-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 21 - Soldotna Bridge

Client Sample ID: **RM 21 - Soldotna Bridge**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869036
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:03
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:06
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:06
Chromium	2.72	J	5.00	2.50	3.75	ug/L	1		07/29/24 20:06
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:06
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:06
Zinc	8.21	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:50

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:50
 Container ID: 1243869036-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:06
 Container ID: 1243869036-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 21 - Soldotna Bridge DUP

Client Sample ID: **RM 21 - Soldotna Bridge DUP**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869037
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:09
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:08
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:08
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:08
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:08
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:08
Zinc	8.23	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:53

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:53
 Container ID: 1243869037-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:08
 Container ID: 1243869037-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 22 - Soldotna Creek

Client Sample ID: **RM 22 - Soldotna Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869038
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:45
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:11
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:11
Chromium	3.62	J	5.00	2.50	3.75	ug/L	1		07/29/24 20:11
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:11
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:11
Zinc	9.52	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:55

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:55
 Container ID: 1243869038-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:11
 Container ID: 1243869038-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 23 - Swiftwater Park

Client Sample ID: **RM 23 - Swiftwater Park**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869039
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:30
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:14
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:14
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:14
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:14
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:14
Zinc	8.54	J	10.0	3.10	7.50	ug/L	1		08/02/24 17:58

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 17:58
 Container ID: 1243869039-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:14
 Container ID: 1243869039-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 30 - Funny River

Client Sample ID: **RM 30 - Funny River**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869040
 Lab Project ID: 1243869

Collection Date: 07/23/24 09:52
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:21
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:21
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:21
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:21
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:21
Zinc	10.0	B	10.0	3.10	7.50	ug/L	1		08/02/24 18:01

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 18:01
 Container ID: 1243869040-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:21
 Container ID: 1243869040-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 30 - Funny River DUP

Client Sample ID: **RM 30 - Funny River DUP**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869041
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:10
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:24
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:24
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:24
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:24
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:24
Zinc	7.71	J	10.0	3.10	7.50	ug/L	1		08/02/24 18:03

Batch Information

Analytical Batch: MMS12377
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/02/24 18:03
 Container ID: 1243869041-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:24
 Container ID: 1243869041-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 12.5 - Pillars

Client Sample ID: **RM 12.5 - Pillars**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869042
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:27
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:27
Chromium	8.33		5.00	2.50	3.75	ug/L	1		07/29/24 20:27
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:27
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:27
Zinc	7.50	U	10.0	3.10	7.50	ug/L	1		08/19/24 11:19

Batch Information

Analytical Batch: MMS12395
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/19/24 11:19
 Container ID: 1243869042-A

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:27
 Container ID: 1243869042-A

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 22 - Soldotna Creek

Client Sample ID: **RM 22 - Soldotna Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869043
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:45
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:29
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:29
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:29
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:29
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:29
Zinc	7.50	U	10.0	3.10	7.50	ug/L	1		08/19/24 11:21

Batch Information

Analytical Batch: MMS12395
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/19/24 11:21
 Container ID: 1243869043-A

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:29
 Container ID: 1243869043-A

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 12.5 - Pillars

Client Sample ID: **RM 12.5 - Pillars**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869044
 Lab Project ID: 1243869

Collection Date: 07/23/24 10:55
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	3.75	U	5.00	1.50	3.75	ug/L	1		07/29/24 20:32
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:32
Chromium	3.75	U	5.00	2.50	3.75	ug/L	1		07/29/24 20:32
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:32
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:32
Zinc	8.74	J	10.0	3.10	7.50	ug/L	1		08/19/24 11:24

Batch Information

Analytical Batch: MMS12395
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/19/24 11:24
 Container ID: 1243869044-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:32
 Container ID: 1243869044-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Results of RM 22 - Soldotna Creek

Client Sample ID: **RM 22 - Soldotna Creek**
 Client Project ID: **Kenai Rive Baseline WQM**
 Lab Sample ID: 1243869045
 Lab Project ID: 1243869

Collection Date: 07/23/24 08:45
 Received Date: 07/23/24 16:55
 Matrix: Water (Surface, Eff., Ground)
 Solids (%):
 Location:

Results by Dissolved Metals by ICP/MS

Parameter	Result	Qual	LOQ/CL	DL	LOD	Units	DF	Allowable Limits	Date Analyzed
Arsenic	4.20	J	5.00	1.50	3.75	ug/L	1		07/29/24 20:35
Cadmium	0.375	U	0.500	0.150	0.375	ug/L	1		07/29/24 20:35
Chromium	4.48	J	5.00	2.50	3.75	ug/L	1		07/29/24 20:35
Copper	2.25	U	3.00	1.00	2.25	ug/L	1		07/29/24 20:35
Lead	1.50	U	2.00	0.500	1.50	ug/L	1		07/29/24 20:35
Zinc	7.99	J	10.0	3.10	7.50	ug/L	1		08/19/24 11:27

Batch Information

Analytical Batch: MMS12395
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 08/19/24 11:27
 Container ID: 1243869045-B

Prep Batch: MXX36834
 Prep Method: E200.2
 Prep Date/Time: 07/31/24 14:54
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Analytical Batch: MMS12371
 Analytical Method: EP200.8
 Analyst: HGS
 Analytical Date/Time: 07/29/24 20:35
 Container ID: 1243869045-B

Prep Batch: MXX36821
 Prep Method: E200.2
 Prep Date/Time: 07/26/24 14:03
 Prep Initial Wt./Vol.: 20 mL
 Prep Extract Vol: 50 mL

Method Blank

Blank ID: MB for HBN 1895761 [MXX/36813]
Blank Lab ID: 1776583

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007

Results by EP200.8

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Calcium	375U	500	150	375	ug/L
Copper	2.25U	3.00	1.00	2.25	ug/L
Iron	188U	250	78.0	188	ug/L
Magnesium	37.5U	50.0	15.0	37.5	ug/L
Zinc	7.50U	10.0	3.10	7.50	ug/L

Batch Information

Analytical Batch: MMS12371
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 7/29/2024 9:42:30PM

Prep Batch: MXX36813
Prep Method: E200.2
Prep Date/Time: 7/25/2024 10:45:00AM
Prep Initial Wt./Vol.: 20 mL
Prep Extract Vol: 50 mL

Print Date: 08/29/2024 4:59:07PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [MXX36813]

Blank Spike Lab ID: 1776584

Date Analyzed: 07/29/2024 21:45

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007

Results by EP200.8

Parameter	Blank Spike (ug/L)			CL
	Spike	Result	Rec (%)	
Calcium	10000	9500	95	(85-115)
Copper	1000	923	92	(85-115)
Iron	5000	4740	95	(85-115)
Magnesium	10000	10300	103	(85-115)
Zinc	1000	908	91	(85-115)

Batch Information

Analytical Batch: **MMS12371**

Analytical Method: **EP200.8**

Instrument: **P7 Agilent 7800**

Analyst: **HGS**

Prep Batch: **MXX36813**

Prep Method: **E200.2**

Prep Date/Time: **07/25/2024 10:45**

Spike Init Wt./Vol.: 5000 ug/L Extract Vol: 50 mL

Dupe Init Wt./Vol.: Extract Vol:

Print Date: 08/29/2024 4:59:12PM



Matrix Spike Summary

Original Sample ID: 1776588
MS Sample ID: 1776589 MS
MSD Sample ID:

Analysis Date: 07/30/2024 12:18
Analysis Date: 07/30/2024 12:20
Analysis Date:
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Calcium	67200	10000	77700	105				70-130		
Copper	1.95J	1000	953	95				70-130		
Iron	465	5000	5330	97				70-130		
Magnesium	15900	10000	26000	101				70-130		
Zinc	57.9	1000	986	93				70-130		

Batch Information

Analytical Batch: MMS12371
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 7/29/2024 10:02:00PM

Prep Batch: MXX36813
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/25/2024 10:45:00AM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Analytical Batch: MMS12372
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 7/30/2024 12:20:53PM

Prep Batch: MXX36813
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/25/2024 10:45:00AM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:13PM

Method Blank

Blank ID: MB for HBN 1895867 [MXX/36820]
Blank Lab ID: 1776876

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024, 1243869028, 1243869029, 1243869030

Results by EP200.8

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Arsenic	3.75U	5.00	1.50	3.75	ug/L
Cadmium	0.375U	0.500	0.150	0.375	ug/L
Calcium	375U	500	150	375	ug/L
Chromium	3.75U	5.00	2.50	3.75	ug/L
Copper	2.25U	3.00	1.00	2.25	ug/L
Iron	188U	250	78.0	188	ug/L
Lead	1.50U	2.00	0.500	1.50	ug/L
Magnesium	37.5U	50.0	15.0	37.5	ug/L
Zinc	56.3*	10.0	3.10	7.50	ug/L

Batch Information

Analytical Batch: MMS12377
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/2/2024 4:10:00PM

Prep Batch: MXX36820
Prep Method: E200.2
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20 mL
Prep Extract Vol: 50 mL

Analytical Batch: MMS12395
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/19/2024 10:16:16AM

Prep Batch: MXX36820
Prep Method: E200.2
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20 mL
Prep Extract Vol: 50 mL

Print Date: 08/29/2024 4:59:15PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [MXX36820]

Blank Spike Lab ID: 1776877

Date Analyzed: 08/19/2024 10:18

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024, 1243869028, 1243869029, 1243869030

Results by EP200.8

Blank Spike (ug/L)				
Parameter	Spike	Result	Rec (%)	CL
Arsenic	1000	860	86	(85-115)
Cadmium	100	91.6	92	(85-115)
Calcium	10000	9510	95	(85-115)
Chromium	400	369	92	(85-115)
Copper	1000	931	93	(85-115)
Iron	5000	4800	96	(85-115)
Lead	1000	919	92	(85-115)
Magnesium	10000	9810	98	(85-115)
Zinc	1000	1010	101	(85-115)

Batch Information

Analytical Batch: **MMS12377**

Analytical Method: **EP200.8**

Instrument: **P7 Agilent 7800**

Analyst: **HGS**

Prep Batch: **MXX36820**

Prep Method: **E200.2**

Prep Date/Time: **07/26/2024 14:03**

Spike Init Wt./Vol.: 5000 ug/L Extract Vol: 50 mL

Dupe Init Wt./Vol.: Extract Vol:

Analytical Batch: **MMS12395**

Analytical Method: **EP200.8**

Instrument: **P7 Agilent 7800**

Analyst: **HGS**

Prep Batch: **MXX36820**

Prep Method: **E200.2**

Prep Date/Time: **07/26/2024 14:03**

Spike Init Wt./Vol.: 1000 ug/L Extract Vol: 50 mL

Dupe Init Wt./Vol.: Extract Vol:

Print Date: 08/29/2024 4:59:19PM

Matrix Spike Summary

Original Sample ID: 1783041
MS Sample ID: 1776879 MS
MSD Sample ID:

Analysis Date: 08/02/2024 17:04
Analysis Date: 08/02/2024 17:12
Analysis Date:
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024, 1243869028, 1243869029

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Arsenic	3.75U	1000	858	86				70-130		
Cadmium	0.375U	100	96.6	97				70-130		
Calcium	13700	10000	24100	104				70-130		
Chromium	3.75U	400	363	91				70-130		
Copper	2.25U	1000	927	93				70-130		
Iron	967	5000	5690	95				70-130		
Lead	1.50U	1000	962	96				70-130		
Magnesium	5630	10000	15400	98				70-130		
Zinc	16.5	1000	966	95				70-130		

Batch Information

Analytical Batch: MMS12377
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/2/2024 5:12:25PM

Prep Batch: MXX36820
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:22PM

Matrix Spike Summary

Original Sample ID: 1783042
MS Sample ID: 1776880 MS
MSD Sample ID:

Analysis Date: 08/02/2024 17:14
Analysis Date: 08/02/2024 17:16
Analysis Date:
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869029, 1243869030

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Arsenic	1.84J	1000	862	86				70-130		
Cadmium	0.375U	100	92.3	92				70-130		
Calcium	36400	10000	46400	100				70-130		
Chromium	3.75U	400	368	92				70-130		
Copper	5.57	1000	903	90				70-130		
Iron	188U	5000	4860	97				70-130		
Lead	1.50U	1000	865	87				70-130		
Magnesium	63100	10000	74500	114				70-130		
Zinc	16.8	1000	930	91				70-130		

Batch Information

Analytical Batch: MMS12377
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/2/2024 5:16:00PM

Prep Batch: MXX36820
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Analytical Batch: MMS12397
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/22/2024 10:27:00AM

Prep Batch: MXX36820
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:22PM

Method Blank

Blank ID: MB for HBN 1895868 [MXX/36821]
Blank Lab ID: 1776881

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869031, 1243869032, 1243869033, 1243869034, 1243869035, 1243869036, 1243869037, 1243869038, 1243869039,
1243869040, 1243869041, 1243869042, 1243869043, 1243869044, 1243869045

Results by EP200.8

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Arsenic	3.75U	5.00	1.50	3.75	ug/L
Cadmium	0.375U	0.500	0.150	0.375	ug/L
Chromium	3.75U	5.00	2.50	3.75	ug/L
Copper	2.25U	3.00	1.00	2.25	ug/L
Lead	1.50U	2.00	0.500	1.50	ug/L

Batch Information

Analytical Batch: MMS12371
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 7/29/2024 7:37:00PM

Prep Batch: MXX36821
Prep Method: E200.2
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20 mL
Prep Extract Vol: 50 mL

Print Date: 08/29/2024 4:59:24PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [MXX36821]

Blank Spike Lab ID: 1776882

Date Analyzed: 07/29/2024 19:39

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869031, 1243869032, 1243869033, 1243869034, 1243869035, 1243869036, 1243869037, 1243869038, 1243869039, 1243869040, 1243869041, 1243869042, 1243869043, 1243869044, 1243869045

Results by EP200.8

Parameter	Blank Spike (ug/L)			CL
	Spike	Result	Rec (%)	
Arsenic	1000	896	90	(85-115)
Cadmium	100	91.1	91	(85-115)
Chromium	400	363	91	(85-115)
Copper	1000	936	94	(85-115)
Lead	1000	987	99	(85-115)

Batch Information

Analytical Batch: MMS12371

Analytical Method: EP200.8

Instrument: P7 Agilent 7800

Analyst: HGS

Prep Batch: MXX36821

Prep Method: E200.2

Prep Date/Time: 07/26/2024 14:03

Spike Init Wt./Vol.: 1000 ug/L Extract Vol: 50 mL

Dupe Init Wt./Vol.: Extract Vol:

Print Date: 08/29/2024 4:59:27PM



Matrix Spike Summary

Original Sample ID: 1243869031

MS Sample ID: 1776884 MS

MSD Sample ID:

Analysis Date: 07/29/2024 19:44

Analysis Date: 07/29/2024 19:51

Analysis Date:

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869031, 1243869032

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Arsenic	3.75U	1000	923	92				70-130		
Cadmium	0.375U	100	94	94				70-130		
Chromium	3.75U	400	375	94				70-130		
Copper	2.25U	1000	955	96				70-130		
Lead	1.50U	1000	994	99				70-130		

Batch Information

Analytical Batch: MMS12371

Analytical Method: EP200.8

Instrument: P7 Agilent 7800

Analyst: HGS

Analytical Date/Time: 7/29/2024 7:51:56PM

Prep Batch: MXX36821

Prep Method: DW Digest for Metals on ICP-MS

Prep Date/Time: 7/26/2024 2:03:00PM

Prep Initial Wt./Vol.: 20.00mL

Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:29PM



Matrix Spike Summary

Original Sample ID: 1243869032
MS Sample ID: 1776885 MS
MSD Sample ID:

Analysis Date: 07/29/2024 19:53
Analysis Date: 07/29/2024 19:56
Analysis Date:
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869032, 1243869033, 1243869034, 1243869035, 1243869036, 1243869037, 1243869038,
1243869039, 1243869040, 1243869041, 1243869042, 1243869043, 1243869044, 1243869045

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Arsenic	3.75U	1000	924	92				70-130		
Cadmium	0.375U	100	91.9	92				70-130		
Chromium	3.75U	400	380	95				70-130		
Copper	1.09J	1000	962	96				70-130		
Lead	1.50U	1000	936	94				70-130		

Batch Information

Analytical Batch: MMS12371
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 7/29/2024 7:56:27PM

Prep Batch: MXX36821
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/26/2024 2:03:00PM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:29PM



Method Blank

Blank ID: MB for HBN 1896339 [MXX/36834]
Blank Lab ID: 1778079

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869031, 1243869032, 1243869033, 1243869034, 1243869035, 1243869036, 1243869037, 1243869038, 1243869039,
1243869040, 1243869041, 1243869042, 1243869043, 1243869044, 1243869045

Results by EP200.8

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Zinc	35.5*	10.0	3.10	7.50	ug/L

Batch Information

Analytical Batch: MMS12395
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/19/2024 11:16:17AM

Prep Batch: MXX36834
Prep Method: E200.2
Prep Date/Time: 7/31/2024 2:54:00PM
Prep Initial Wt./Vol.: 20 mL
Prep Extract Vol: 50 mL

Print Date: 08/29/2024 4:59:31PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [MXX36834]

Blank Spike Lab ID: 1778080

Date Analyzed: 08/02/2024 17:24

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869031, 1243869032, 1243869033, 1243869034, 1243869035, 1243869036, 1243869037, 1243869038, 1243869039, 1243869040, 1243869041, 1243869042, 1243869043, 1243869044, 1243869045

Results by EP200.8

Parameter	Blank Spike (ug/L)			CL
	Spike	Result	Rec (%)	
Zinc	1000	958	96	(85-115)

Batch Information

Analytical Batch: MMS12377

Analytical Method: EP200.8

Instrument: P7 Agilent 7800

Analyst: HGS

Prep Batch: MXX36834

Prep Method: E200.2

Prep Date/Time: 07/31/2024 14:54

Spike Init Wt./Vol.: 1000 ug/L Extract Vol: 50 mL

Dupe Init Wt./Vol.: Extract Vol:

Print Date: 08/29/2024 4:59:35PM



Matrix Spike Summary

Original Sample ID: 1778141
MS Sample ID: 1778143 MS
MSD Sample ID:

Analysis Date: 08/02/2024 17:41
Analysis Date: 08/02/2024 17:44
Analysis Date:
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869031, 1243869032, 1243869033, 1243869034, 1243869035

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Zinc	9.41J	1000	956	95				70-130		

Batch Information

Analytical Batch: MMS12377
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/2/2024 5:44:15PM

Prep Batch: MXX36834
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/31/2024 2:54:00PM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:37PM

Matrix Spike Summary

Original Sample ID: 1778142
MS Sample ID: 1778144 MS
MSD Sample ID:

Analysis Date: 08/02/2024 17:46
Analysis Date: 08/02/2024 17:48
Analysis Date:
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869035, 1243869036, 1243869037, 1243869038, 1243869039, 1243869040, 1243869041,
1243869042, 1243869043, 1243869044, 1243869045

Results by EP200.8

Parameter	Sample	Matrix Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Zinc	6.96J	1000	939	93				70-130		

Batch Information

Analytical Batch: MMS12377
Analytical Method: EP200.8
Instrument: P7 Agilent 7800
Analyst: HGS
Analytical Date/Time: 8/2/2024 5:48:46PM

Prep Batch: MXX36834
Prep Method: DW Digest for Metals on ICP-MS
Prep Date/Time: 7/31/2024 2:54:00PM
Prep Initial Wt./Vol.: 20.00mL
Prep Extract Vol: 50.00mL

Print Date: 08/29/2024 4:59:37PM

Method Blank

Blank ID: MB for HBN 1896756 [VXX/41596]
Blank Lab ID: 1778856

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869002, 1243869003, 1243869017, 1243869018, 1243869025, 1243869026, 1243869027

Results by SW8260D

Parameter	Results	LOQ/CL	DL	LOD	Units
Benzene	0.300U	0.400	0.120	0.300	ug/L
Ethylbenzene	0.750U	1.00	0.310	0.750	ug/L
o-Xylene	0.750U	1.00	0.310	0.750	ug/L
P & M -Xylene	1.50U	2.00	0.620	1.50	ug/L
Toluene	0.750U	1.00	0.310	0.750	ug/L
Xylenes (total)	2.25U	3.00	1.00	2.25	ug/L

Surrogates

1,2-Dichloroethane-D4 (surr)	104	81-118		0	%
4-Bromofluorobenzene (surr)	99.6	85-114		0	%
Toluene-d8 (surr)	102	89-112		0	%

Batch Information

Analytical Batch: VMS23497
Analytical Method: SW8260D
Instrument: Agilent 7890-75MS
Analyst: MZ
Analytical Date/Time: 8/3/2024 4:11:00AM

Prep Batch: VXX41596
Prep Method: SW5030B
Prep Date/Time: 8/2/2024 6:00:00AM
Prep Initial Wt./Vol.: 5 mL
Prep Extract Vol: 5 mL

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [VXX41596]
 Blank Spike Lab ID: 1778857
 Date Analyzed: 08/03/2024 04:26

Spike Duplicate ID: LCSD for HBN 1243869 [VXX41596]
 Spike Duplicate Lab ID: 1778858
 Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869002, 1243869003, 1243869017, 1243869018, 1243869025, 1243869026, 1243869027

Results by SW8260D

Parameter	Blank Spike (ug/L)			Spike Duplicate (ug/L)			CL	RPD (%)	RPD CL
	Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Benzene	30	28.3	94	30	28.7	96	(79-120)	1.50	(< 20)
Ethylbenzene	30	30.1	100	30	29.7	99	(79-121)	1.30	(< 20)
o-Xylene	30	30.4	101	30	29.7	99	(78-122)	2.20	(< 20)
P & M -Xylene	60	61.9	103	60	61.7	103	(80-121)	0.37	(< 20)
Toluene	30	28.9	96	30	28.8	96	(80-121)	0.17	(< 20)
Xylenes (total)	90	92.3	103	90	91.4	102	(79-121)	0.98	(< 20)
Surrogates									
1,2-Dichloroethane-D4 (surr)	30		97	30		95	(81-118)	1.40	
4-Bromofluorobenzene (surr)	30		99	30		100	(85-114)	1.10	
Toluene-d8 (surr)	30		103	30		103	(89-112)	0.07	

Batch Information

Analytical Batch: VMS23497
 Analytical Method: SW8260D
 Instrument: Agilent 7890-75MS
 Analyst: MZ

Prep Batch: VXX41596
 Prep Method: SW5030B
 Prep Date/Time: 08/02/2024 06:00
 Spike Init Wt./Vol.: 30 ug/L Extract Vol: 5 mL
 Dupe Init Wt./Vol.: 30 ug/L Extract Vol: 5 mL

Print Date: 08/29/2024 4:59:42PM

Method Blank

Blank ID: MB for HBN 1895900 (WFI/3132)
Blank Lab ID: 1777049

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007

Results by SM21 4500NO3-F

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Nitrate-N	0.150U	0.200	0.0500	0.150	mg/L
Nitrite-N	0.150U	0.200	0.0500	0.150	mg/L
Total Nitrate/Nitrite-N	0.150U	0.200	0.0500	0.150	mg/L

Batch Information

Analytical Batch: WFI3132
Analytical Method: SM21 4500NO3-F
Instrument: Astoria segmented flow
Analyst: AJP
Analytical Date/Time: 7/26/2024 1:21:44PM

Print Date: 08/29/2024 4:59:45PM

Method Blank

Blank ID: MB for HBN 1895900 (WFI/3132)
Blank Lab ID: 1777051

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007, 1243869008, 1243869009,
1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016, 1243869017, 1243869018,
1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500NO3-F

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Nitrate-N	0.150U	0.200	0.0500	0.150	mg/L
Nitrite-N	0.150U	0.200	0.0500	0.150	mg/L
Total Nitrate/Nitrite-N	0.150U	0.200	0.0500	0.150	mg/L

Batch Information

Analytical Batch: WFI3132
Analytical Method: SM21 4500NO3-F
Instrument: Astoria segmented flow
Analyst: AJP
Analytical Date/Time: 7/26/2024 2:07:14PM

Print Date: 08/29/2024 4:59:45PM

Method Blank

Blank ID: MB for HBN 1895900 (WFI/3132)

Matrix: Water (Surface, Eff., Ground)

Blank Lab ID: 1777053

QC for Samples:

1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500NO3-F

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Nitrate-N	0.150U	0.200	0.0500	0.150	mg/L
Nitrite-N	0.150U	0.200	0.0500	0.150	mg/L
Total Nitrate/Nitrite-N	0.150U	0.200	0.0500	0.150	mg/L

Batch Information

Analytical Batch: WFI3132

Analytical Method: SM21 4500NO3-F

Instrument: Astoria segmented flow

Analyst: AJP

Analytical Date/Time: 7/26/2024 2:52:44PM

Print Date: 08/29/2024 4:59:45PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [WFI3132]

Blank Spike Lab ID: 1777048

Date Analyzed: 07/26/2024 13:19

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007

Results by SM21 4500NO3-F

Parameter	Blank Spike (mg/L)			CL
	<u>Spike</u>	<u>Result</u>	<u>Rec (%)</u>	
Nitrate-N	2.5	2.69	108	(70-130)
Nitrite-N	2.5	2.69	107	(90-110)
Total Nitrate/Nitrite-N	5	5.38	108	(90-110)

Batch Information

Analytical Batch: **WFI3132**

Analytical Method: **SM21 4500NO3-F**

Instrument: **Astoria segmented flow**

Analyst: **AJP**

Print Date: 08/29/2024 4:59:49PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [WFI3132]

Blank Spike Lab ID: 1777050

Date Analyzed: 07/26/2024 14:05

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007, 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021,

Results by SM21 4500NO3-F

Blank Spike (mg/L)				
Parameter	Spike	Result	Rec (%)	CL
Nitrate-N	2.5	2.71	108	(70-130)
Nitrite-N	2.5	2.61	105	(90-110)
Total Nitrate/Nitrite-N	5	5.33	107	(90-110)

Batch Information

Analytical Batch: WFI3132

Analytical Method: SM21 4500NO3-F

Instrument: Astoria segmented flow

Analyst: AJP

Print Date: 08/29/2024 4:59:49PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [WFI3132]

Blank Spike Lab ID: 1777052

Date Analyzed: 07/26/2024 14:50

Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500NO3-F

Blank Spike (mg/L)				
Parameter	Spike	Result	Rec (%)	CL
Nitrate-N	2.5	2.52	101	(70-130)
Nitrite-N	2.5	2.65	106	(90-110)
Total Nitrate/Nitrite-N	5	5.17	103	(90-110)

Batch Information

Analytical Batch: WFI3132

Analytical Method: SM21 4500NO3-F

Instrument: Astoria segmented flow

Analyst: AJP

Print Date: 08/29/2024 4:59:49PM



Matrix Spike Summary

Original Sample ID: 1243755004
MS Sample ID: 1777021 MS
MSD Sample ID: 1777022 MSD

Analysis Date: 07/26/2024 12:39
Analysis Date: 07/26/2024 12:41
Analysis Date: 07/26/2024 12:43
Matrix: Water (Surface, Eff., Ground)

QC for Samples:

Results by SM21 4500NO3-F

Parameter	Sample	Matrix Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Nitrate/Nitrite-N	0.200U	5.00	4.85	97	5.00	5.27	105	90-110	8.30	(< 25)

Batch Information

Analytical Batch: WFI3132
Analytical Method: SM21 4500NO3-F
Instrument: Astoria segmented flow
Analyst: AJP
Analytical Date/Time: 7/26/2024 12:41:00PM

Print Date: 08/29/2024 4:59:51PM



Matrix Spike Summary

Original Sample ID: 1243833001
MS Sample ID: 1777023 MS
MSD Sample ID: 1777024 MSD

Analysis Date: 07/26/2024 13:25
Analysis Date: 07/26/2024 13:26
Analysis Date: 07/26/2024 13:28
Matrix: Drinking Water

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007

Results by SM21 4500NO3-F

Parameter	Sample	Matrix Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Nitrate/Nitrite-N	0.200U	5.00	4.92	98	5.00	4.95	99	90-110	0.66	(< 25)

Batch Information

Analytical Batch: WFI3132
Analytical Method: SM21 4500NO3-F
Instrument: Astoria segmented flow
Analyst: AJP
Analytical Date/Time: 7/26/2024 1:26:00PM

Print Date: 08/29/2024 4:59:51PM



Matrix Spike Summary

Original Sample ID: 1243862001
MS Sample ID: 1777025 MS
MSD Sample ID: 1777026 MSD

Analysis Date: 07/26/2024 14:10
Analysis Date: 07/26/2024 14:12
Analysis Date: 07/26/2024 14:14
Matrix: Drinking Water

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007, 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021.

Results by SM21 4500NO3-F

Parameter	Sample	Matrix Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Nitrate/Nitrite-N	0.200U	5.00	5.7	114 *	5.00	5.57	111 *	90-110	2.40	(< 25)

Batch Information

Analytical Batch: WFI3132
Analytical Method: SM21 4500NO3-F
Instrument: Astoria segmented flow
Analyst: AJP
Analytical Date/Time: 7/26/2024 2:12:00PM

Print Date: 08/29/2024 4:59:51PM

Matrix Spike Summary

Original Sample ID: 1243876001
MS Sample ID: 1777027 MS
MSD Sample ID: 1777028 MSD

Analysis Date: 07/26/2024 14:56
Analysis Date: 07/26/2024 14:57
Analysis Date: 07/26/2024 14:59
Matrix: Drinking Water

QC for Samples: 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016, 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500NO3-F

Parameter	Sample	Matrix Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Nitrate/Nitrite-N	0.896	5.00	6.43	111 *	5.00	6.70	116 *	90-110	4.10	(< 25)

Batch Information

Analytical Batch: WFI3132
Analytical Method: SM21 4500NO3-F
Instrument: Astoria segmented flow
Analyst: AJP
Analytical Date/Time: 7/26/2024 2:57:00PM

Print Date: 08/29/2024 4:59:51PM



Method Blank

Blank ID: MB for HBN 1896475 [WXX/15350]
Blank Lab ID: 1778406

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007, 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016

Results by SM21 4500P-B,E

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Total Phosphorus	0.0300U	0.0400	0.0120	0.0300	mg/L

Batch Information

Analytical Batch: WDA5837
Analytical Method: SM21 4500P-B,E
Instrument: Discrete Analyzer 2
Analyst: AJP
Analytical Date/Time: 8/1/2024 1:06:21PM

Prep Batch: WXX15350
Prep Method: SM21 4500P-B,E
Prep Date/Time: 7/31/2024 12:14:00PM
Prep Initial Wt./Vol.: 25 mL
Prep Extract Vol: 25 mL

Print Date: 08/29/2024 4:59:53PM

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [WXX15350]
 Blank Spike Lab ID: 1778407
 Date Analyzed: 08/01/2024 13:07

Spike Duplicate ID: LCSD for HBN 1243869 [WXX15350]
 Spike Duplicate Lab ID: 1778408
 Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007, 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016

Results by SM21 4500P-B,E

Parameter	Blank Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
	Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Phosphorus	0.2	0.191	96	0.2	0.189	94	(75-125)	1.20	(< 25)

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Instrument: Discrete Analyzer 2
 Analyst: AJP

Prep Batch: WXX15350
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/2024 12:14
 Spike Init Wt./Vol.: 0.2 mg/L Extract Vol: 25 mL
 Dupe Init Wt./Vol.: 0.2 mg/L Extract Vol: 25 mL

Print Date: 08/29/2024 4:59:56PM



Matrix Spike Summary

Original Sample ID: 1243869001
MS Sample ID: 1778409 MS
MSD Sample ID: 1778410 MSD

Analysis Date: 08/01/2024 13:12
Analysis Date: 08/01/2024 13:13
Analysis Date: 08/01/2024 13:14
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869001, 1243869002, 1243869003, 1243869004, 1243869005, 1243869006, 1243869007, 1243869008, 1243869009, 1243869010, 1243869011, 1243869012, 1243869013, 1243869014, 1243869015, 1243869016

Results by SM21 4500P-B,E

Parameter	Sample	Matrix Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Phosphorus	0.0300U	0.200	.209	105	0.200	0.207	103	75-125	1.10	(< 7)

Batch Information

Analytical Batch: WDA5837
Analytical Method: SM21 4500P-B,E
Instrument: Discrete Analyzer 2
Analyst: AJP
Analytical Date/Time: 8/1/2024 1:13:15PM

Prep Batch: WXX15350
Prep Method: Total Phosphorus (W) Ext.
Prep Date/Time: 7/31/2024 12:14:00PM
Prep Initial Wt./Vol.: 25.00mL
Prep Extract Vol: 25.00mL

Print Date: 08/29/2024 4:59:58PM



Method Blank

Blank ID: MB for HBN 1896476 [WXX/15351]
Blank Lab ID: 1778411

Matrix: Water (Surface, Eff., Ground)

QC for Samples:

1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500P-B,E

<u>Parameter</u>	<u>Results</u>	<u>LOQ/CL</u>	<u>DL</u>	<u>LOD</u>	<u>Units</u>
Total Phosphorus	0.0300U	0.0400	0.0120	0.0300	mg/L

Batch Information

Analytical Batch: WDA5837
Analytical Method: SM21 4500P-B,E
Instrument: Discrete Analyzer 2
Analyst: AJP
Analytical Date/Time: 8/1/2024 1:33:49PM

Prep Batch: WXX15351
Prep Method: SM21 4500P-B,E
Prep Date/Time: 7/31/2024 12:14:00PM
Prep Initial Wt./Vol.: 25 mL
Prep Extract Vol: 25 mL

Print Date: 08/29/2024 5:00:00PM

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200 West Potter Drive Anchorage, AK 95518
t 907.562.2343 f 907.561.5301 www.us.sgs.com

Member of SGS Group

Blank Spike Summary

Blank Spike ID: LCS for HBN 1243869 [WXX15351]
 Blank Spike Lab ID: 1778412
 Date Analyzed: 08/01/2024 13:34

Spike Duplicate ID: LCSD for HBN 1243869 [WXX15351]
 Spike Duplicate Lab ID: 1778413
 Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500P-B,E

Parameter	Blank Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
	Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Phosphorus	0.2	0.177	88	0.2	0.179	90	(75-125)	1.50	(< 25)

Batch Information

Analytical Batch: WDA5837
 Analytical Method: SM21 4500P-B,E
 Instrument: Discrete Analyzer 2
 Analyst: AJP

Prep Batch: WXX15351
 Prep Method: SM21 4500P-B,E
 Prep Date/Time: 07/31/2024 12:14
 Spike Init Wt./Vol.: 0.2 mg/L Extract Vol: 25 mL
 Dupe Init Wt./Vol.: 0.2 mg/L Extract Vol: 25 mL

Print Date: 08/29/2024 5:00:04PM



Matrix Spike Summary

Original Sample ID: 1243869017
MS Sample ID: 1778414 MS
MSD Sample ID: 1778415 MSD

Analysis Date: 08/01/2024 13:36
Analysis Date: 08/01/2024 13:37
Analysis Date: 08/01/2024 13:38
Matrix: Water (Surface, Eff., Ground)

QC for Samples: 1243869017, 1243869018, 1243869019, 1243869020, 1243869021, 1243869022, 1243869023, 1243869024

Results by SM21 4500P-B,E

Parameter	Sample	Matrix Spike (mg/L)			Spike Duplicate (mg/L)			CL	RPD (%)	RPD CL
		Spike	Result	Rec (%)	Spike	Result	Rec (%)			
Total Phosphorus	0.0300U	0.200	.196	98	0.200	0.195	98	75-125	0.10	(< 7)

Batch Information

Analytical Batch: WDA5837
Analytical Method: SM21 4500P-B,E
Instrument: Discrete Analyzer 2
Analyst: AJP
Analytical Date/Time: 8/1/2024 1:37:43PM

Prep Batch: WXX15351
Prep Method: Total Phosphorus (W) Ext.
Prep Date/Time: 7/31/2024 12:14:00PM
Prep Initial Wt./Vol.: 25.00mL
Prep Extract Vol: 25.00mL

Print Date: 08/29/2024 5:00:06PM



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Working together for healthy watersheds on the Kenai Peninsula

July 1, 2024

To:

Curtis Whisman
SGS North America, Inc
200 W Potter Dr
Anchorage, AK 99518

Hello Curtis,

Included with this email is the kit request for our Summer 2024 Kenai River Baseline Water Quality Sampling project. The materials can be mailed to our address in Soldotna listed above. We will be collecting samples on July 23rd this year, and sending them back to your lab in Anchorage via Grant Aviation that same afternoon.

The kit request is similar to last year's kit request for Summer 2023. A few notes for specific analyses follow below:

Dissolved Metals Analyses

Dissolved metals samples (EPA 200.8 analysis), including field blanks and duplicates, will be filtered in the SGS lab.

Parameters to be analyzed in the 200.8 analyses include:

- Arsenic
- Cadmium
- Chromium
- Copper
- Lead
- Zinc

Total Metals Analyses

Total metals (EPA 200.8 analysis) samples are collected at all sites throughout the project. We request analyses for total metals samples for five parameters at sites from RM 31 (River Mile 31) and downstream to RM 0, including:

- Calcium, total
- Iron, total
- Magnesium, total
- Copper, total



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-
- Zinc, total

At sites upstream of RM 31, we request a smaller subset of total metals results, including:

- Calcium, total
- Iron, total
- Magnesium, total

Other Notes

- We are collecting metals field blanks at two sites, which will be analyzed for a.) total metals and b.) dissolved metals. These additions are reflected in the kit request numbers.
- BTEX trip blanks: we request a total of three (3) BTEX trip blank kits. When results from BTEX trip blanks are reported, we request that they are associated with the site name(s) written on the label on the outer box of the kit.
- The project QAPP can be accessed on our website at <https://www.kenaiwatershed.org/news-media/qapp-revisions-completed-2023/>.

Thank you again for your time and support with this project, and please don't hesitate to contact me with any questions.

Sincerely,

A handwritten signature in black ink, appearing to read 'Ben Meyer', with a stylized flourish at the end.

Benjamin Meyer, Water Quality Coordinator

ben@kenaiwatershed.org

(907) 232-0280



SGS North America Inc.
CHAIN OF CUSTODY RECORD

1243869



Profile # 383466 CSW

CLIENT: Kenai Watershed Forum						Instructions: Sections 1 - 5 must be filled out. Omissions may delay the onset of analysis.										Page <u>1</u> of <u>3</u>									
CONTACT: Benjamin Meyer						PHONE #: 907-232-0280		Section 3		Preservative															
PROJECT NAME: Kenai River Baseline Water Quality Monitoring						PROJECT/ PWSID/ PERMIT#:		#		Analysis*										NOTE: *The following analyses require specific method and/or compound list: BTEX, Metals, PFAS					
REPORTS TO: Benjamin Meyer						E-MAIL: ben@kenaiwatershed.org		Comp																	
INVOICE TO: Kenai Watershed Forum						QUOTE #:		Grab																	
P.O. #:								MI (Multi-incremental)																	
RESERVED for lab use		SAMPLE IDENTIFICATION		DATE mm/dd/yy		TIME HH:MM		MATRIX/ MATRIX CODE				Total NO ₃ /NO ₂ (SM21 4900NCS-F), Total P(SM4500)		Total Metals (200.8)		Dissolved Metals (200.8)		BTEX (8260C)						REMARKS/LOC ID	
1AB		RM 0 - No Name Creek 28AB		7/23/2024		8:59		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
2AE		RM 1.5 - Kenai City Dock 29AB		7/23/2024		10:00		water		6		x		x		x		x						Dissolved metals samples are unfiltered and unpreserved	
3AE		RM 6.5 - Cunningham Park 30AB		7/23/2024		9:20		water		6		x		x		x		x						Dissolved metals samples are unfiltered and unpreserved	
4AB		RM 10 - Beaver Creek 31AB		7/23/2024		10:05		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
5AB		RM 10.1 - Kenai River 32AB		7/23/2024		10:32		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
6AD		RM 12.5 - Pillars 33AB		7/23/2024		10:55		water		5		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
7AB		RM 18 - Poacher's Cove 34AB		7/23/2024		11:37		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
8AB		RM 19 - Slikok Creek 35AB		7/23/2024		10:32		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
9AB		RM 21 - Soldotna Bridge 36AB		7/23/2024		10:03		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
10AB		RM 21 - Soldotna Bridge - DUP 37AB		7/23/2024		10:09		water		3		x		x		x								Dissolved metals samples are unfiltered and unpreserved	
Relinquished By: (1) Ben Meyer						Date 7/23/2024		Time		Received By:		Section 4		DOD Project? Yes (No)		Data Deliverable Requirements: Please include Electronic Data Delivery files.									
Relinquished By: (2)						Date		Time		Received By:		Cooler ID:		Requested Turnaround Time and/or Special Instructions: Filter Dissolved metals samples at SGS. Please provide all Electronic Data Delivery (EDD) files on SGS Enengage when available.											
Relinquished By: (3)						Date		Time		Received By:		Temp Blank °C: 3.4 D57		or Ambient []		Chain of Custody Seal: (Circle) INTACT <u>15</u> BROKEN ABSENT									
Relinquished By: (4)						Date 7/23/24		Time 11:55		Received For Laboratory By: luhw		Delivery Method: Hand Delivery [] Commerical Delivery <u>Hand</u>													

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Section 1						Instructions: Sections 1 - 5 must be filled out. Omissions may delay the onset of analysis.														Page <u>2</u> of <u>3</u>			
CLIENT: Kenai Watershed Forum						Section 3		Preservative															
CONTACT: Benjamin Meyer PHONE #: 907-232-0280						# CONTAINER	Comp Grab MI (Multi-incremental)	Analysis*														NOTE: *The following analyses require specific method and/or compound list: BTEX, Metals, PFAS	
PROJECT NAME: Kenai River Baseline Water Quality Monitoring PROJECT/ PWSID/ PERMIT#:																							
REPORTS TO: Benjamin Meyer E-MAIL: ben@kenaiwatershed.org Profile #:																							
INVOICE TO: Kenai Watershed Forum QUOTE #: P.O. #:																							
RESERVED for lab use		SAMPLE IDENTIFICATION		DATE mm/dd/yy	TIME HH:MM	MATRIX/ MATRIX CODE															REMARKS/LOC ID		
11AD		RM 22 - Soldotna Creek 38AB		7/23/2024	8:45	water	5		x	x	x	(43A) (45AB)											
12AB		RM 23 - Swiftwater Park 39AB		7/23/2024	9:30	water	3		x	x	x											Dissolved metals samples are unfiltered and unpreserved	
13AB		RM 30 - Funny River 40AB		7/23/2024	9:52	water	3		x	x	x											Dissolved metals samples are unfiltered and unpreserved	
14AB		RM 30 - Funny River - DUP 41AB		7/23/2024	10:10	water	3		x	x	x											Dissolved metals samples are unfiltered and unpreserved	
15AB		RM 31 - Morgan's Landing		7/23/2024	11:23	water	2		x	x													
16A		RM 36 - Moose River		7/23/2024	12:40	water	1		x	EA													
17AC		RM 40 - Bing's Landing		7/23/2024	10:55	water	5		x	x		x											
18AC		RM 43 - Upstream of Dow Island		7/23/2024	10:25	water	5		x	x		x											Please include site name on Trip Blank sample name results
19AB		RM 44 - Mouth of Killey River		7/23/2024	10:00	water	2		x	x													
20AB		RM 50 - Skilak Lake Outflow		7/23/2024	9:05	water	2		x	x													
Relinquished By: (1)						Date 7/23/2024	Time	Received By:		Section 4		DOD Project? Yes No		Data Deliverable Requirements: Please include Electronic Data Delivery files.									
Relinquished By: (2)						Date	Time	Received By:		Requested Turnaround Time and/or Special Instructions: Filter Dissolved metals samples at SGS. Please provide all Electronic Data Delivery (EDD) files on SGS Enegage when available.													
Relinquished By: (3)						Date	Time	Received By:		Temp Blank °C: _____ or Ambient []													
Relinquished By: (4)						Date 7/23/24	Time 1655	Received For Laboratory By: [Signature]		Chain of Custody Seal: (Circle) INTACT 1F BROKEN ABSENT Delivery Method: Hand Delivery [] Commerical Delivery [X]													

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CHAIN OF CUSTODY RECORD

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CLIENT: Kenai Watershed Forum					Instructions: Sections 1 - 5 must be filled out. Omissions may delay the onset of analysis.										Page <u>3</u> of <u>3</u>						
CONTACT: Benjamin Meyer					PHONE #: 907-232-0280		Section 3		Preservative												
PROJECT NAME: Kenai River Baseline Water Quality Monitoring					PROJECT/ PWSID/ PERMIT#:		# CONTAINERS	Comp Grab MI (Multi-incremental)	Analysis*										NOTE: *The following analyses require specific method and/or compound list: BTEX, Metals, PFAS		
REPORTS TO: Benjamin Meyer					E-MAIL: ben@kenaiwatershed.org														REMARKS/LOC ID		
INVOICE TO: Kenai Watershed Forum					QUOTE #: P.O. #:																
RESERVED for lab use		SAMPLE IDENTIFICATION		DATE mm/dd/yy	TIME HH:MM	MATRIX/ MATRIX CODE			Total NO3/NO2(SM21 4500NO3-F), Total P(SM4500)	Total Metals (200.8)	BTEX										
21AB	RM 70 - Jim's Landing	7/23/2024	11:50	water	2			x	x												
22AB	RM 74 - Russian River	7/23/2024	11:00	water	2			x	x												
23AB	RM 82 - Kenai Lake Bridge	7/23/2024	8:40	water	2			x	x												
24AB	RM 79.5 - Juneau Creek	7/23/2024	9:40	water	2			x	x												
25AC	RM 1.5 - Trip Blank	7/23/2024	10:00	water still	3							X									
26AC				water still	3							X									
	RM 40, 43, 44, 50 - Trip Blank	7/23/2024	10:25	water still	3							X									
27AC	RM 6.5 - Trip Blank	7/23/2024	9:20	water still	3							X									
Relinquished By: (1)		Date 7/23/2024	Time	Received By:		Section 4		DOD Project? Yes No		Data Deliverable Requirements: Please include Electronic Data Delivery files.											
Relinquished By: (2)		Date	Time	Received By:		Cooler ID:		Requested Turnaround Time and/or Special Instructions: Please provide all Electronic Data Delivery (EDD) files on SGS Enegage when available.													
Relinquished By: (3)		Date	Time	Received By:		Temp Blank °C:		Chain of Custody Seal: (Circle) INTACT BROKEN ABSENT													
Relinquished By: (4)		Date 7/23/24	Time 11:55	Received For Laboratory By: [Signature]		or Ambient []		Delivery Method: Hand Delivery [] Commerical Delivery [X]													

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SAMPLE RECEIPT FORM

Project Manager Completion				
Was all necessary information recorded on the COC upon receipt? (temperature, COC seals, etc.?)	☒ Yes	☐ No	☐ N/A	
Was temperature between 0-6 °C?	☒ Yes	☐ No	☐ N/A	If "No", are the samples either exempt* or sampled <8 hours prior to receipt?
Were all analyses received within holding time*?	☒ Yes	☐ No	☐ N/A	
Was a method specified for each analysis, where applicable? If no, please note correct methods.	☒ Yes	☐ No	☐ N/A	
Are compound lists specified, where applicable? For project specific or special compound lists please note correct analysis code.	☒ Yes	☐ No	☐ N/A	TB's associated w/ specific samples
If rush was requested by the client, was the requested TAT approved?	☐ Yes	☐ No	☒ N/A	If "NO", what is the approved TAT?
If SEDD Deliverables are required, were Location ID's and an NPDL Number provided?	☐ Yes	☐ No	☒ N/A	If "NO", contact client for information.
Sample Login Completion				
Do ID's on sample containers match COC?	☒ Yes	☐ No	☐ N/A	
If provided on containers, do dates/times collected match COC?	☒ Yes	☐ No	☐ N/A	Note: If times differ <1 hr., record details below and login per COC.
Were all sample containers received in good condition?	☒ Yes	☐ No	☐ N/A	
Were proper containers (type/mass/volume/preservative) received for all samples? *See form F-083 "Sample Guide"	☒ Yes	☐ No	☐ N/A	Note: If 200.8/6020 Total Metals are received unpreserved, preserve and note HNO3 lot here: If 200.8/6020 Dissolved Metals are received unpreserved, log in for LABFILTER and do not preserve. For all non-metals methods, inform Project Manager.
Were Trip Blanks (VOC, GRO, Low-Level Hg, etc.) received with samples, where applicable*?	☒ Yes	☐ No	☐ N/A	
Were all VOA vials free of headspace >6mm?	☒ Yes	☐ No	☐ N/A	
Were all soil VOA samples received field extracted with Methanol?	☐ Yes	☐ No	☒ N/A	
Did all soil VOA samples have an accompanying unpreserved container for % solids?	☐ Yes	☐ No	☒ N/A	
If special handling is required, were containers labelled appropriately? e.g. MI/ISM, foreign soils, lab filter, Ref Lab, limited volume	☐ Yes	☐ No	☒ N/A	
For Rush/Short Holding time, was the lab notified?	☐ Yes	☐ No	☒ N/A	
For any question answered "NO", was the Project Manager notified?	☐ Yes	☐ No	☒ N/A	PM Initials:
Was Peer Review of sample numbering/labelling completed?	☒ Yes	☐ No	☐ N/A	Reviewer Initials: JGG
Additional Notes/Clarification where Applicable, including resolution of "No" answers when a change order is not attached:				

AIRBILL 14037860

I hereby declare that the goods contained herein do not contain dangerous goods.

Signed.....

Date

Grant Aviation
6420 Kulis Dr. Anchorage, AK 99502
Phone: 1 (888) 359-4726



Freephone
Em:
Web: htt

1243869



FREIGHT DETAILS

FROM/TO: Kenai -> Anchorage International

Receiver: SGS North America
907-272-0349

Sender: Kenai Water shed Fourm
907-232-0280

Accepted:

:55 PM
:00 PM

Description & Comment	Quan.	Wgt.	Handle Fee	Hazmat Fee	Total
Standard Freight	1	53	-	-	\$39.91
Total Tax:					\$2.49
Total Payments made:					\$42.40
Total Unpaid:					\$0.00

Received in good condition by:

CUSTOMER COPY

AIRBILL 14037860

I hereby declare that the goods contained herein do not contain dangerous goods.

Signed.....

Date

Grant Aviation
6420 Kulis Dr. Anchorage, AK 99502
Phone: 1 (888) 359-4726
Freephone: 1 (888) 359-4726
Email: res@flygrant.com
Web: http://www.flygrant.com/



FREIGHT DETAILS

FROM/TO: Kenai -> Anchorage International

Receiver: SGS North America
907-272-0349

Sender: Kenai Water shed Fourm
907-232-0280

Flight Departs: Jul 23 24 2:55 PM

Accepted: Tue, Jul 23 24 2:39:00 PM

Description & Comment	Quan.	Wgt.	Handle Fee	Hazmat Fee	Total
Standard Freight	1	53	-	-	\$39.91
TAX: Federal Excise Tax					\$2.49
Total Payments made:					\$42.40
Total Unpaid:					\$0.00

TERMS AND CONDITIONS

Consignemnt Note Text

Alert Expeditors Inc.

#433991

Citywide Delivery • 440-3351
8421 Flamingo Drive • Anchorage, Alaska 99502



Date

7-23-24

From

Kona Watershed

To

565 Labs Ave

Collect ☐

Prepay ☐

Advance Charges ☐

Job #

E/A

PO#

Grant / 1057860

Samples

1243869



Shipped Signature

Total Charge

Received By:

Sample Containers and Preservatives

<u>Container Id</u>	<u>Preservative</u>	<u>Container Condition</u>	<u>Container Id</u>	<u>Preservative</u>	<u>Container Condition</u>
1243869001-A	H2SO4 to pH < 2	OK	1243869020-A	H2SO4 to pH < 2	OK
1243869001-B	HNO3 to pH < 2	OK	1243869020-B	HNO3 to pH < 2	OK
1243869002-A	H2SO4 to pH < 2	OK	1243869021-A	H2SO4 to pH < 2	OK
1243869002-B	HNO3 to pH < 2	OK	1243869021-B	HNO3 to pH < 2	OK
1243869002-C	HCL to pH < 2	OK	1243869022-A	H2SO4 to pH < 2	OK
1243869002-D	HCL to pH < 2	OK	1243869022-B	HNO3 to pH < 2	OK
1243869002-E	HCL to pH < 2	OK	1243869023-A	H2SO4 to pH < 2	OK
1243869003-A	H2SO4 to pH < 2	OK	1243869023-B	HNO3 to pH < 2	OK
1243869003-B	HNO3 to pH < 2	OK	1243869024-A	H2SO4 to pH < 2	OK
1243869003-C	HCL to pH < 2	OK	1243869024-B	HNO3 to pH < 2	OK
1243869003-D	HCL to pH < 2	OK	1243869025-A	HCL to pH < 2	OK
1243869003-E	HCL to pH < 2	OK	1243869025-B	HCL to pH < 2	OK
1243869004-A	H2SO4 to pH < 2	OK	1243869025-C	HCL to pH < 2	OK
1243869004-B	HNO3 to pH < 2	OK	1243869026-A	HCL to pH < 2	OK
1243869005-A	H2SO4 to pH < 2	OK	1243869026-B	HCL to pH < 2	OK
1243869005-B	HNO3 to pH < 2	OK	1243869026-C	HCL to pH < 2	OK
1243869006-A	H2SO4 to pH < 2	OK	1243869027-A	HCL to pH < 2	OK
1243869006-B	HNO3 to pH < 2	OK	1243869027-B	HCL to pH < 2	OK
1243869007-A	H2SO4 to pH < 2	OK	1243869027-C	HCL to pH < 2	OK
1243869007-B	HNO3 to pH < 2	OK	1243869028-A	No Preservative Required	OK
1243869008-A	H2SO4 to pH < 2	OK	1243869028-B	No Preservative Required	OK
1243869008-B	HNO3 to pH < 2	OK	1243869029-A	No Preservative Required	OK
1243869009-A	H2SO4 to pH < 2	OK	1243869029-B	No Preservative Required	OK
1243869009-B	HNO3 to pH < 2	OK	1243869030-A	No Preservative Required	OK
1243869010-A	H2SO4 to pH < 2	OK	1243869030-B	No Preservative Required	OK
1243869010-B	HNO3 to pH < 2	OK	1243869031-A	No Preservative Required	OK
1243869011-A	H2SO4 to pH < 2	OK	1243869031-B	No Preservative Required	OK
1243869011-B	HNO3 to pH < 2	OK	1243869032-A	No Preservative Required	OK
1243869012-A	H2SO4 to pH < 2	OK	1243869032-B	No Preservative Required	OK
1243869012-B	HNO3 to pH < 2	OK	1243869033-A	No Preservative Required	OK
1243869013-A	H2SO4 to pH < 2	OK	1243869033-B	No Preservative Required	OK
1243869013-B	HNO3 to pH < 2	OK	1243869034-A	No Preservative Required	OK
1243869014-A	H2SO4 to pH < 2	OK	1243869034-B	No Preservative Required	OK
1243869014-B	HNO3 to pH < 2	OK	1243869035-A	No Preservative Required	OK
1243869015-A	H2SO4 to pH < 2	OK	1243869035-B	No Preservative Required	OK
1243869015-B	HNO3 to pH < 2	OK	1243869036-A	No Preservative Required	OK
1243869016-A	H2SO4 to pH < 2	OK	1243869036-B	No Preservative Required	OK
1243869017-A	H2SO4 to pH < 2	OK	1243869037-A	No Preservative Required	OK
1243869017-B	HNO3 to pH < 2	OK	1243869037-B	No Preservative Required	OK
1243869017-C	HCL to pH < 2	OK	1243869038-A	No Preservative Required	OK
1243869017-D	HCL to pH < 2	OK	1243869038-B	No Preservative Required	OK
1243869017-E	HCL to pH < 2	OK	1243869039-A	No Preservative Required	OK
1243869018-A	H2SO4 to pH < 2	OK	1243869039-B	No Preservative Required	OK
1243869018-B	HNO3 to pH < 2	OK	1243869040-A	No Preservative Required	OK
1243869018-C	HCL to pH < 2	OK	1243869040-B	No Preservative Required	OK
1243869018-D	HCL to pH < 2	OK	1243869041-A	No Preservative Required	OK
1243869018-E	HCL to pH < 2	OK	1243869041-B	No Preservative Required	OK
1243869019-A	H2SO4 to pH < 2	OK	1243869042-A	HNO3 to pH < 2	OK
1243869019-B	HNO3 to pH < 2	OK	1243869043-A	HNO3 to pH < 2	OK

<u>Container Id</u>	<u>Preservative</u>	<u>Container Condition</u>	<u>Container Id</u>	<u>Preservative</u>	<u>Container Condition</u>
1243869044-A	No Preservative Required	OK			
1243869044-B	No Preservative Required	OK			
1243869045-A	No Preservative Required	OK			
1243869045-B	No Preservative Required	OK			

Container Condition Glossary

Containers for bacteriological, low level mercury and VOA vials are not opened prior to analysis and will be assigned condition code OK unless evidence indicates than an inappropriate container was submitted.

OK - The container was received at an acceptable pH for the analysis requested.

BU - The container was received with headspace greater than 6mm.

DM - The container was received damaged.

FR - The container was received frozen and not usable for Bacteria or BOD analyses.

IC - The container provided for microbiology analysis was not a laboratory-supplied, pre-sterilized container and therefore was not suitable for analysis.

NC- The container provided was not preserved or was under-preserved. The method does not allow for additional preservative added after collection.

PA - The container was received outside of the acceptable pH for the analysis requested. Preservative was added upon receipt and the container is now at the correct pH. See the Sample Receipt Form for details on the amount and lot # of the preservative added.

PH - The container was received outside of the acceptable pH for the analysis requested. Preservative was added upon receipt, but was insufficient to bring the container to the correct pH for the analysis requested. See the Sample Receipt Form for details on the amount and lot # of the preservative added.

QN - Insufficient sample quantity provided.